INTRODUCTION.

These Instructions might equally be titled An Introduction to Machine Knitting on the Dubied Hand Machine. They were written to acquaint the beginner gradually and methodically with her machine. The order of the subjects follows the teaching given in our knitting schools, so that the beginner may study our instructions paragraph by paragraph, never turning a page before she has fully understood it and carried out on her machine the instructions it gives.

As soon as she has grasped the general working of her machine and knows a first kind of stitch, the knitter puts her knowledge into practice by making a simple article: a scarf. She then learns other kinds of stitch, which allow her to knit in turn a pullover, a skirt and a sleeved jacket. Next come fancy stitches and various kinds of pattern, which are used to make a cardigan. Finally the knitter completes her training by making stockings and socks, which already call for a certain skill.

The knitter then knows enough to fend for herself and to reproduce with ease the varied articles that appear each month in the Revue du Tricotage published by us, or to create new articles as her fancy takes her.

The information given relates to the DUBIED machine of type MM, 60 cm. wide and of gauge 32 or 36. Those who own a Dubied hand machine of another type, M, MR or CS, can nevertheless use these instructions in full, for all our hand machines work on the same principle. On pages 59 and 60 we give some details relating to the M, MR and CS machines, and we ask those who own one of these machines to kindly refer to those pages during their training.

DUBIED

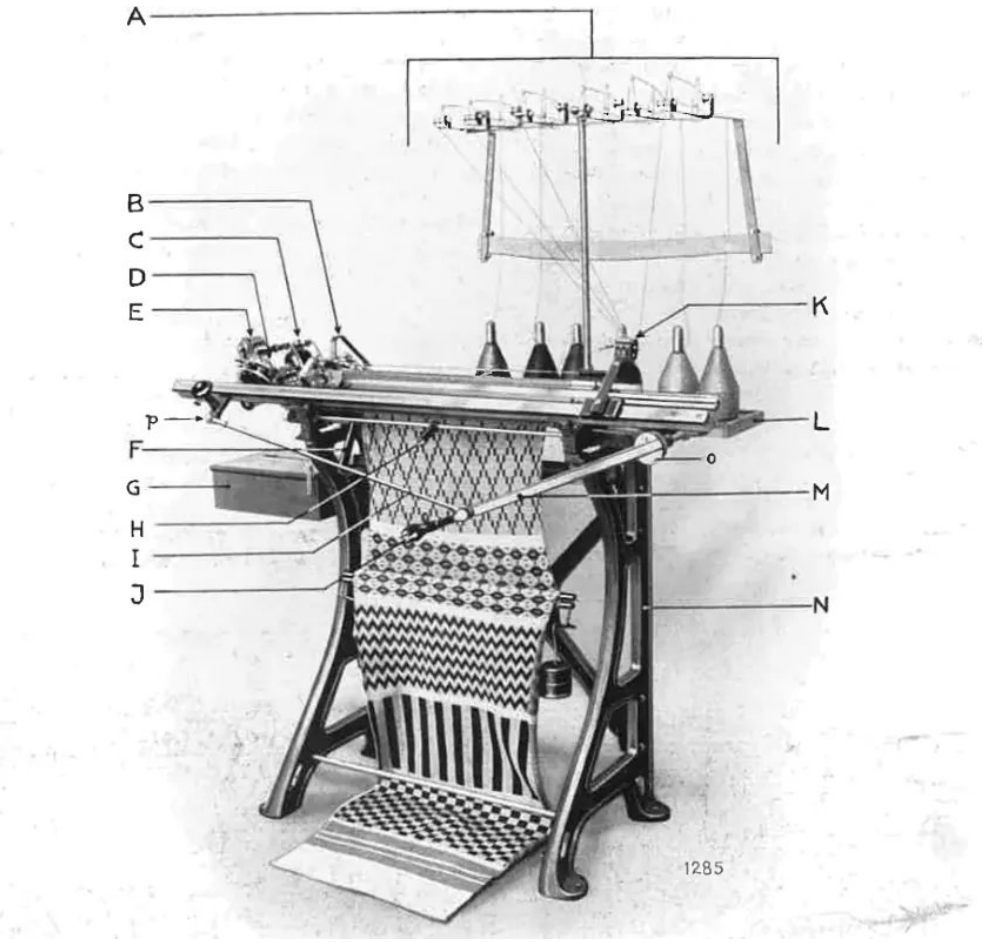


Fig. 1. DUBIED machine. Type MM with Super-8 and 6-colour push-button yarn changer.

KEY.

A = Tension mast
B = Yarn guide
C = Carriage
D = Super-8
E = Pattern indicator
F = Racking handle
G = Accessory box
H = Bed-bed-lowering handle
I = Connecting rod
J = Crank handle grip
K = Push-button yarn changer
L = Table leaf
M = Crank
N = Cast-iron stand

Main parts of the knitting machine.

The flat knitting machine with latch needles was built for the first time in Europe, at Couvet (Switzerland) in 1867, by Edouard Dubied senior. The modern machine of the Dubied firm consists of a roof-shaped cast-iron *frame* that supports the two *needle beds*, also simply called *beds*. These are two *steel plates* milled with grooves in which *the needles* slide.

Above the needle beds runs *the carriage*, guided in its to-and-fro movement by *slide bars*. The carriage bears mechanisms called *cams or locks*, which move the needles. At each passage of the carriage along the needle beds the needles rise and fall, and each forms a *stitch*.

The needles (see *a*, fig. 3) end at one end in a swelling or *butt d*, on which the cam boxes act; at the other end in a *hook b* and a hinged *tongue* or *latch c*, which serves to form the stitch.

Latch needles work in the same way as ordinary crochet hooks: with the latch open, that is, folded back on the body of the needle, a yarn is caught with the hook and the needle is moved forward so that this yarn slides behind the latch; a new yarn is caught in the hook as the needle is drawn back, the first yarn closes the latch, slides over it and over the hook, and drops astride the strand of the 2nd yarn held by the hook, thus forming a stitch (see fig. 3).

In the Dubied knitting machine the needles are arranged side by side and form stitches in succession from the same yarn; this yarn travels along the row of needles alternately from right to left and from left to right. In this way one obtains a strip of knitting whose width depends on the number of needles used. The fineness of the knitting depends mainly on the spacing of the needles.

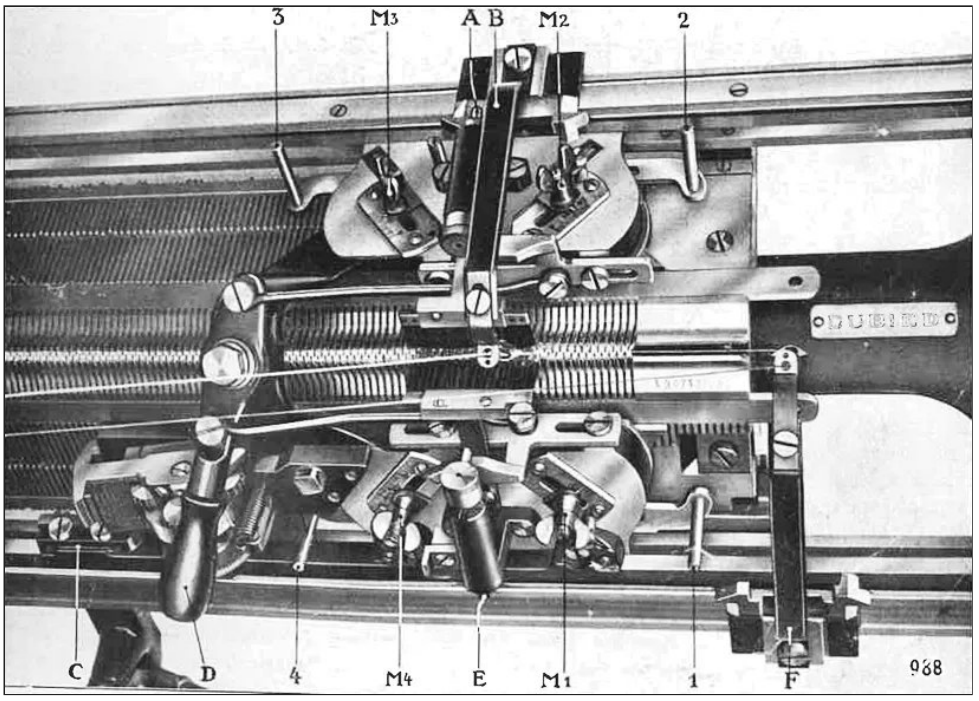


Fig. 2. Carriage of the machine.

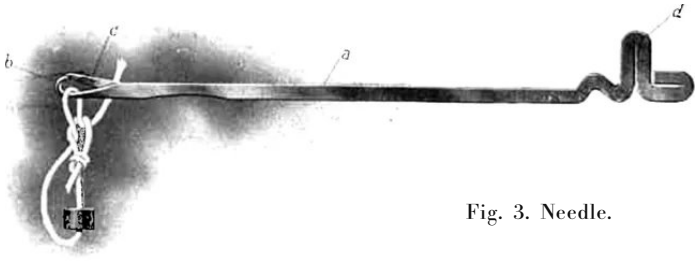


Fig. 3. Needle.

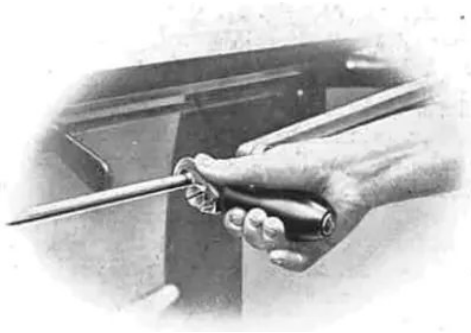


Fig. 4. Crank stop on the left.

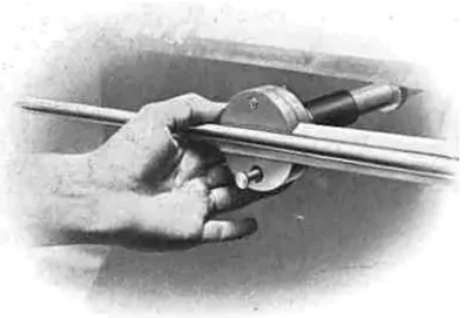


Fig. 5. Crank stop on the right.

The carriage.

The *carriage* is driven by a *crank* to which it is joined by a *connecting rod*. This crank can be shortened, that is, it can be slid from left to right in the crank head, after loosening the screw *o*, fig. 1, with a key and pulling out the small stop plunger next to the screw head.

It is advisable to shorten the crank when knitting narrow articles that need only a limited field of needles. With the shortened crank one will work faster and more easily.

The carriage consists of a *bridge* extended on each side by two *plates* containing the *cam box* mechanisms and the *cams*. The carriage is driven by means of the *crank*. When the carriage is stopped at one end of the machine, it tends to run back, drawn by the weight of the crank; to avoid this drawback, stops have been placed at both ends of the travel.

To stop the carriage on the left, engage on the connecting rod the *hook* (fig. 4) located beside the crank handle; to stop it on the right, engage under the connecting rod the small *plunger* (fig. 5) that passes through the crank head.

The guiding of the carriage by the slide bars is extremely precise thanks to an *extension C* (fig. 2), which bears on the slide through an adjustable *gib* screwed to the left end of the extension.

The cam box.

The cam box is the name given to the cams and their support as a whole.

The *raising cams A* (fig. 6) make the needles rise. They are put in work by pulling the cam-control pins 1, 2, 3 and 4 (fig. 2) towards the outside of the carriage, and out of work by pushing the pins inwards.

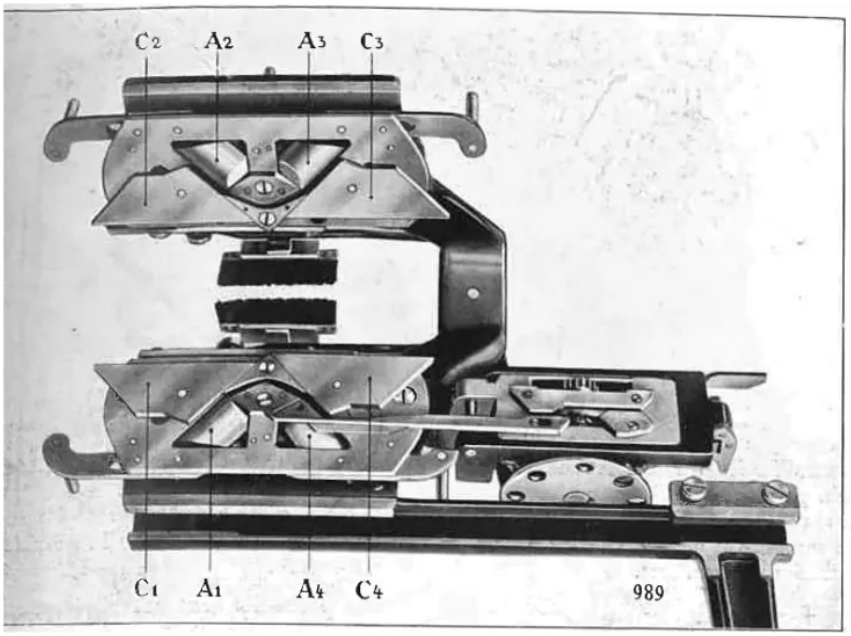


Fig. 6. Carriage seen from below.

The *stitch cams C* (fig. 6) make the needles descend and determine the length of the stitches. The setting of each stitch cam is shown by the *graduated scale* (fig. 7), along which a pointer *c* moves. On the rod bearing the *pointer* is a *stitch-cam knob* a, which serves to fix the stitch cam.

The operation of placing the stitch cams at a given position is called *"setting the knobs"*. This setting is given, for the various kinds of stitch,

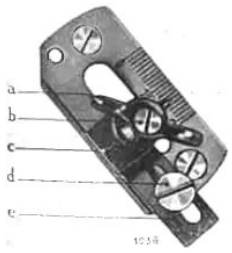


Fig. 7. Stitch-cam knob.

on a special sheet supplied with the machine, called the *"Knob Setting Sheet"*.

To avoid breaking stitches when using fragile yarns, one can work with *free stitch cams (free knobs)*, that is, set the position of the knobs without tightening them.

For this the knob stops are used, as follows (fig. 7):

bring the pointer *c* to the desired position, tighten the knob *a*, loosen the screw *d* a little, press the stop *e* against the pointer *c*, and firmly retighten the screw *d*;

unscrew the knob 2 or 3 turns, screw in the lock screw *b* by hand, then screw the knob back to hold it on the rod of the stitch cam. The stitch cam will then be free. To fix the stitch cams again, release the knob, unscrew the screw *b* by hand, then screw down the knob.

It is not essential to lock the knobs if only a few rows are knitted with the stitch cams free.

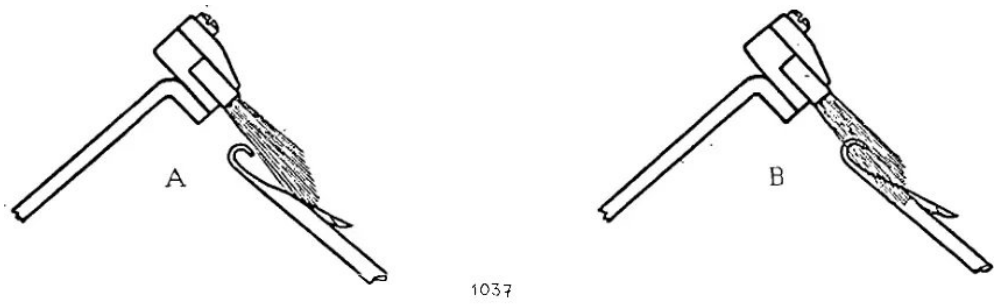


Fig. 8. Brushes.

The brushes.

The brushes fixed to the carriage open the needle latches, so that the needle hooks are sure to take the yarn. They are placed so as to skim the needles, with the bevel facing upwards.

Diagram *A* (fig. 8) shows a correct position of the brushes, diagram *B* a faulty position, which causes rapid wear of the brushes.

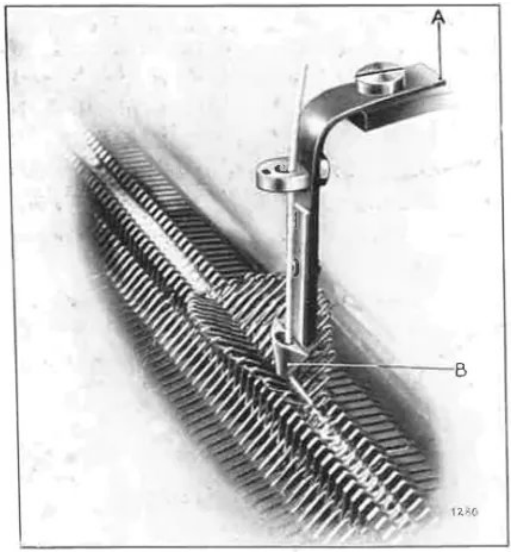


Fig. 9. Yarn-carrier tip.

The yarn changer (see fig. 2).

The yarn changer is the mechanism used to bring the yarn to the needles. Its essential parts are the *yarn carriers F* and *B*, which slide one on the front of the machine, the other on the rear.

The yarn carriers are driven by the carriage by means of the *plungers A* and *E*. These are controlled by the *lever D*. Pushing this lever to the left puts the rear yarn carrier in work and the front yarn carrier out of work. Pushing it to the right gives the opposite result.

The change of yarn carrier is always made with the carriage to the right of the machine.

The *yarn-carrier tip B* (fig. 9) is the very hard tempered steel piece at the end of the yarn carrier *A*. It is pierced with a hole through which the yarn passes. So that the yarn runs very smoothly, the parts of the yarn-carrier tip it touches are carefully polished and rounded.

The counter.

A small counter, reading up to 1000 cycles and driven by the crank shaft, shows the number of cycles of the carriage. To make good use of the counter, one need only set it to zero when beginning work.

Preparatory work.

Unpacking and installing a machine.

Take off the lid, remove the accessories and the various parts fixed to the inner walls of the crate, then unscrew from the outside the wooden crosspieces that wedge the machine.

Stand the crate on two chairs and unscrew from underneath (from the outside) the 4 screws holding the machine to the bottom of the crate.

Finally lift out the machine, holding it by the bars near the point where they meet the frame, and not by the ends of the bars, which the weight of the machine could bend.

This unpacking must in all cases be done easily and without forcing anything, so as not to risk damaging the machine.

Place the machine on the cast-iron stand (*N*, fig. 1) and fix it with the same 4 screws that held it to the bottom of the crate.

If the cast-iron stand has arrived dismantled, assemble the two legs, joining them with the crosspiece or crosspieces provided. Tightening the knobs is enough to obtain a rigid assembly. Fix the wooden shelf to the back of the machine and the box to the left leg.

Join the connecting rod to the carriage with the screw *p*, and screw on the crank handle *J* (fig. 1).

Slide the yarn carriers onto their bar, yarn carrier No. 1 on the front, yarn carrier No. 2 on the rear (see *F* and *B*, fig. 2).

Screw onto the wooden shelf the socket *B*, fig. 10, through which passes the column of the tension mast, which is clamped in this socket. On the column place the crosspiece that receives the tension units (see *A*, fig. 10) and screw these to the crosspiece.

If the machine was ordered without the cast-iron stand, choose for it a perfectly level table with an absolutely flat surface, or the machine will work badly.

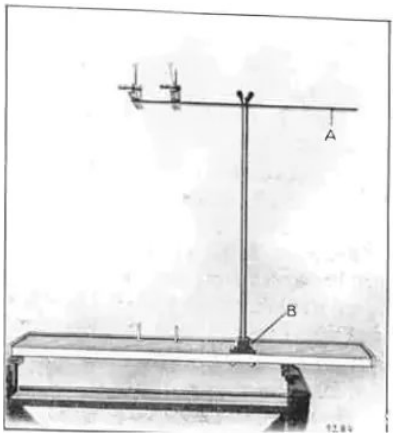


Fig. 10. Cone and tension stand.

Fix the two clamping brackets to the machine with the screws that held it in the crate; set them against the edge of the table and screw them firmly, using either bolts or large wood screws.

Take great care that the machine is horizontal and that both brackets lie perfectly against the table.

If this is not the case, small wedges of wood or cardboard must be inserted between the table and the brackets at the points required.

Then fix the tension mast; for this, fix the socket B, fig. 10, to the table with wood screws.

Cleaning and greasing.

The needle beds, bars and cam boxes must be cleaned with rags of the anti-rust petroleum jelly that covers them.

The only really effective way of cleaning the needles is to take them out of the needle bed and rub them with a brush or a rag soaked in petroleum benzine.

Likewise clean the grooves with a brush and petroleum benzine, and put the needles back in place, taking care not to damage the latches.

Then lightly oil the rubbing parts.

Preparing the cones.

The yarn must be wound on *cones* in such a way that it can unwind evenly, without any resistance. The *swift* and the *cone winder* are fixed at some distance from each other on the edge of a table (see fig. 11).

Place the skein on the swift, fit the cone firmly on the spindle of the winder and wind the yarn onto the cone, starting from the bottom and moving gradually towards the tip, to obtain a true cone.

Fig. 12 shows a well-made cone, fig. 13 a faulty one.

While winding, move the yarn to and fro so that the turns cross one another. If this precaution is neglected, the turns lie parallel on top of one another and, if the skein stops suddenly, the yarn is pulled tight and sinks into the wound part. Unwinding is then faulty.

If the skein has knots, do not let them pass but replace them with the weaver's knot or the reef knot, see fig. 14 and 15. Above all, never wax the knots.

Wax the yarn, especially wool of inferior or harsh quality, by passing it as it is wound over a piece of special Dubied paraffin wax held in the left hand guiding the yarn (see fig. 11).

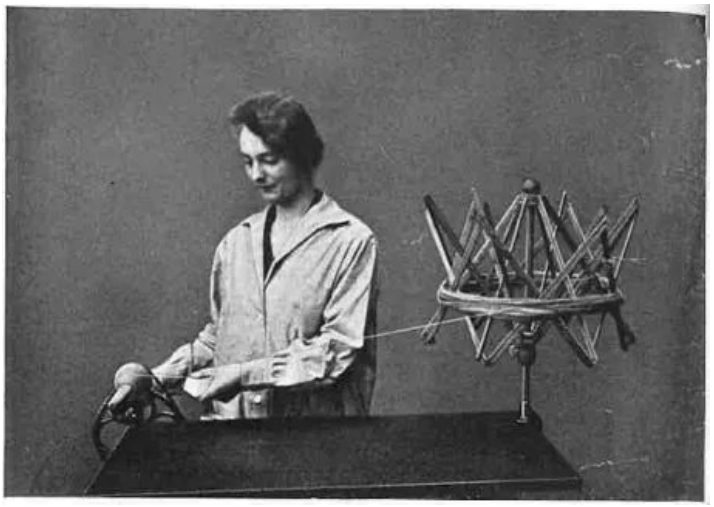


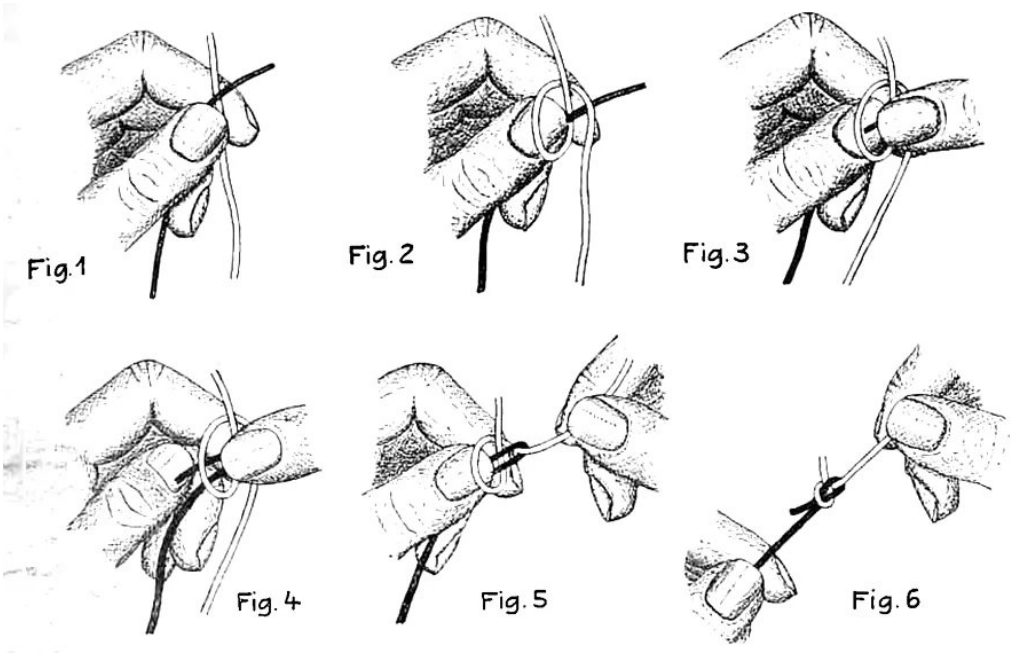
Fig. 11. Winding.



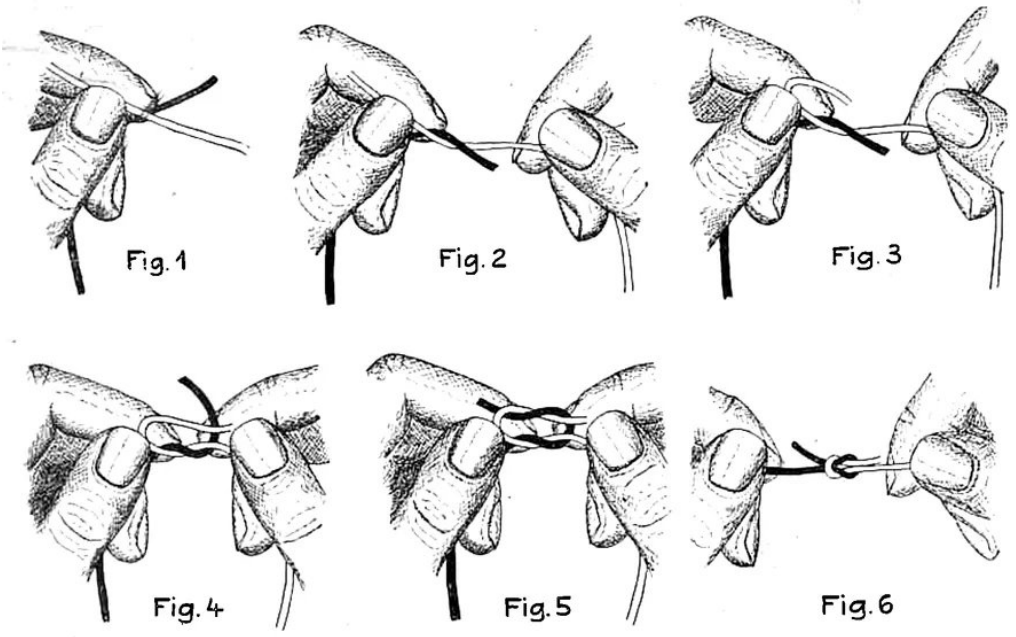
Fig. 12. Well-made cone.



Fig. 13. Badly made cone.



Weaver's knot



Reef knot.

Figures 14 and 15.

Threading the yarn.

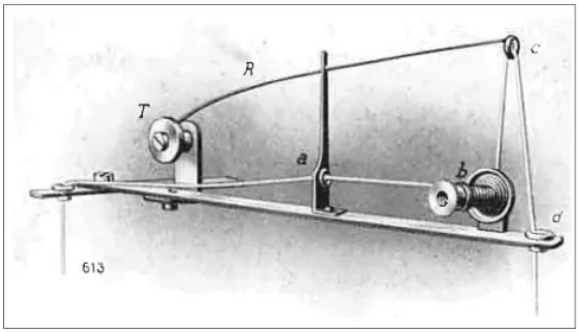


Fig. 16. A tension unit.

From the cone the yarn rises vertically to an eyelet on the tension mast crosspiece (fig. 16), passes from there to *the eyelet a*, enters the *tension discs b* between the two washers, which are parted to bring it into the notch of the spindle, goes through *the eyelet c* of the *tension arm R* and finally *the eyelet d*, from which it goes directly to the *yarn-carrier tip*, which brings it to the needles.

The tension is set as follows with the *tension discs b* and the *drum T*:

turn the drum *T* so as to bring the tension arm *R* roughly upright, slackening the yarn so that it does not act on the arm,

take hold of the yarn below the eyelet *d* and pull it evenly. When the eyelet *c* of the arm comes roughly level with the tension discs *b*, the yarn should begin to unwind from the cone. If the yarn unwinds sooner, tighten the tension discs a little; if on the contrary it does not unwind easily, the tension is too strong, and the tension discs must be loosened.

As a general rule, try to keep the tension to a minimum, so as to avoid any needless strain on the yarn. However, the coarser the gauge, the thicker the yarns used and the stronger the tension must be.

Casting on, welt, 1 × 1 rib.

Stand in front of the machine, a little to the left, so as to be able to handle the crank easily.

Raise the required number of needles, for example 40 on the rear and 40 on the front, by pushing the safety springs fully upwards, paying no attention to the upper stop notch, which can be passed without harm since the needle returns to its normal position when the carriage passes. Take care to lower the unused needles as far as possible.

When putting the needles in work, count them with the graduations on the needle-retaining strip at the bottom of the front bed.

Distribute half of the total number of needles to be put in work on each side of the *0* in the middle of the strip.

The needles must be arranged as follows:

I I I I I ..... I I I I I Rear
I I I I I ..... I I I I I Front

Note carefully that the last needle on the left is on the front bed and the last needle on the right on the rear bed.

Pull out the 4 pins 1 to 4, fig. 2.

Set the knobs M 1 to M 4 (fig. 2) to the following divisions:

for machines of gauge 36 (No. 7):

13 13
×
13 13

for machines of gauge 32 (No. 8)

12 12
×
12 12

Knobs 3 and 4 are set with the carriage on the left, and knobs 1 and 2 with the carriage on the right.

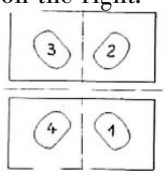


Fig. 17.

To understand clearly what these figures mean, see the diagram opposite (Position of the knobs).

As a general rule, the rear yarn carrier is used for the ground yarn, that is, the yarn making up the greater part of an article. The front yarn carrier is used for the colour yarn, that is, the yarn intended to form stripes or patterns.

Place the stop blocks of the yarn carriers far enough to the left, 5 cm. beyond the last needle in work.

Place the cone on the shelf, pass the yarn through the tension unit and the yarn carrier as shown in fig. 16, draw it under the machine between the two needle beds and pinch it under a weight. (Hang the weight holders on the rod below the needle beds at the front of the machine.)

Push in pins 1 and 3 (fig. 2).

Pulling on the comb with the left hand, knit two passes (one cycle), that is, move the carriage to the right, then bring it back to the left. (Tubular knitting.)

Hook on to the cast-on comb the weight holders, loaded according to the width of the knitting (see table page 14).

Pull out pins 1 and 3.

Work the carriage evenly: at each pass it produces, simultaneously, a row of stitches on the front needles and a row of stitches on the rear needles. This is what is called 1 × 1 rib.

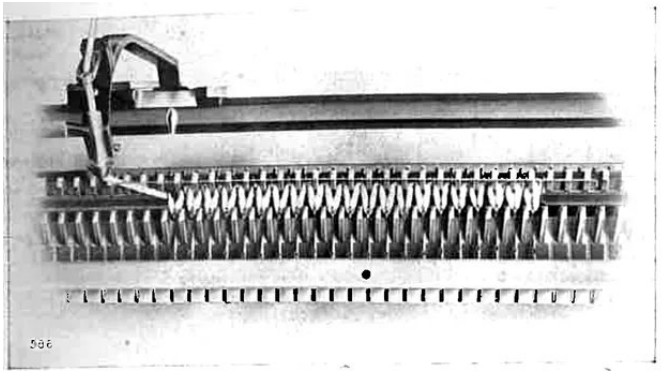


Fig. 18. Casting on.

To take the knitting off the machine:

Stop the carriage on the right, take off the weights hanging from the comb, free the carriage from the two yarn carriers, take the cast-on comb in the left hand and, pulling it slightly, move the carriage to the left without a yarn carrier. The knitting will thus come off the machine. Take care to break the yarn and pinch it under a weight, so that it is ready for the next piece of work.

Bring the carriage to the right, push the yarn changer handle *D*, fig. 2, to the left and move the carriage slowly to the left, checking that the rear yarn carrier is taken along; the two rows of needles will catch the yarn, forming a net (see fig. 18).

Take the cast-on comb in the left hand and insert it from below between the two needle beds (fig. 19) until all its eyelets are above the net.

Insert the comb wire from right to left, making sure that the yarn coming from the yarn carrier is also covered (see fig. 18).

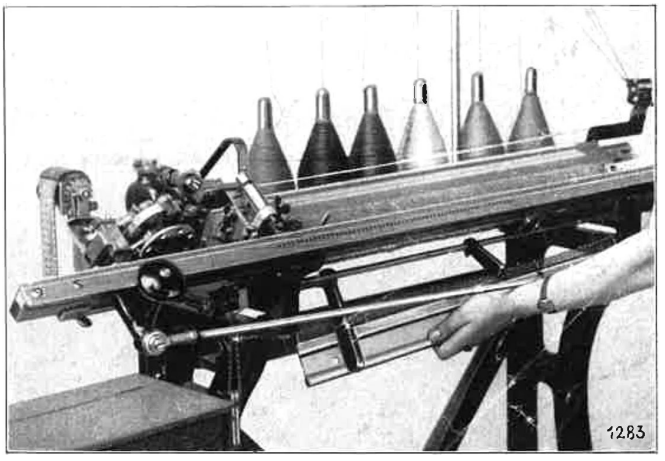


Fig. 19. Inserting the comb.

Amount of weight to hang from the knitting.

In general, note:

that, except for narrow knitting, it is better to hang weights at each end of the comb rather than in the middle only;

that it is essential to hang the same amount of weight on each side of the comb, so that it stays horizontal;

that tubular knitting and 1 × 1 rib knitted with auxiliary stitch cams (see page 61) need more weight than other kinds of knitting.

In the table below, the letters P.M.G. stand respectively for small, medium and large. The entry 1 M and 1 P means, for example, that one medium weight and one small weight are to be put on each weight holder.

Table of the weights to hang from the knitting.

Width in cm. of the field of needles in work . . .	1 weight holder			2 weight holders					3 weight holders			
	2	5	10	20	30	40	50	60	70	80	90	100
Weights per weight holder. . . .	1 P	1 P	1 M	1 P	1 M	1 G	1 G 1 P	1 G 1 M	1 G	1 G 1 P	1 G 1 M	1 G 1 M 1 P

The amount of weight depends on the material used. If the stitches form badly, increase the weight.

SCARF in 1 × 1 rib.

Scarf with stripes at the ends. (1 × 1 rib.)

200 grams of beige wool 20/2. Brown stripe.

Put about 120 needles in work.

After the cast-on and the 3 tubular cycles, once pins 1 and 3 have been pulled out, knit for example:

12½ cycles in wool of ground colour,
8 " " " " the other colour,
2 " " " " ground colour,
6 " " " " the other colour,
2 " " " " ground colour,
4 " " " " the other colour,
1 cycle " " " ground colour,
2 cycles " " " the other colour,
200 cycles in wool of ground colour,
2 " " " " the other colour,
1 cycle " " " ground colour,
4 cycles " " " the other colour,
2 " " " " ground colour,
6 " " " " the other colour,
2 " " " " ground colour,
8 " " " " the other colour,
12½ " " " " ground colour.

While knitting this scarf, use the edge-weight loop (see next page).





Fig. 20 and 21. Fitting the edge-weight loop.

Using the Edge-Weight Loop.

When knitting a piece of some length, there comes a moment when the weights hanging from the comb touch the floor. They must then be raised to maintain the pull on the knitting. This can be done in two ways:

1° Roll the knitting round the comb by turning it, so that it rises towards the needle beds, and place the hooks of the weight holders on the back of the comb.

This method is suitable if the length of the knitted piece does not exceed one metre; beyond that, the roll of knitting round the comb becomes too bulky and awkward.

2° Take off the weights and the comb and use the edge-weight loop, as follows:

a) Pass the knitting down through the loop, so as to have the 2 tubes in front (fig. 20).

b) Pass the knitting up between the two tubes and let it fall in front of the first tube (fig. 21).

c) Place the edge-weight loop as high as possible, horizontally, tip the weight bar over and hang the weights on it (see fig. 1).

To raise the edge-weight loop when the weights reach the floor, take hold of the selvedges of the knitting in front of the edge-weight loop and pull the knitting so as to slide the edge-weight loop towards the needle beds.

Chain cast-off.

Finish the scarf with the chain cast-off as follows:

Transfer all the stitches from the front needles to the rear needles in this way:

open the latches of all the needles in work,

with the transfer tool, take the stitch of the first needle of the front bed on the left (see page 30), move the tool a little to the right to place it exactly opposite the first needle of the rear bed, lift the tool to lengthen the stitch a little, taking care not to let it slip,

push the rear needle up a little so that its hook enters the stitch held by the tool, hook the stitch and withdraw the tool.

Continue in the same way up to the right-hand selvedge.

Knit one row with all 4 pins pulled out.

Release the stitches of the front bed; for this, push all the front needles up, then bring them back down.

Break the yarn so that it no longer pulls on the stitches, push in pins 1 and 4 and carefully take the knitting off the machine.

Make the chain cast-off with a machine needle, as follows:

hook the needle into the first, then into the second stitch, until the tip of the latch lies between the two stitches; withdraw the needle, and the second stitch will pass through the first; hook the third stitch, pass the second behind the latch, and so on. At the end, pass the end of the yarn through the last stitch.

The article will be edged as shown in fig. 22.

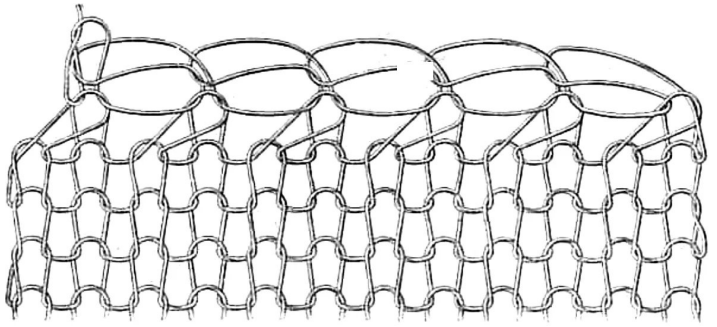


Fig. 22. Chain cast-off.

Once the scarf is finished, attach fringes at both ends.

Tubular knitting.

For tubular knitting, make the cast-on as for 1 × 1 rib, but with the following knob positions:

Gauge 36 (No. 7)		Gauge 32 (No. 8)	
3 / 16	2 / 10	3 / 14	2 / 10
×		×	
4 / 10	1 / 16	4 / 10	1 / 14

Go on knitting with pins 1 and 3 kept closed throughout.

Knitting on one needle bed (Jersey).

For knitting on one needle bed (the rear bed, for example), transfer the stitches of the front bed to the rear bed with the transfer tool (see page 30).

Put the needles of the front bed out of work, or else close pins 1 and 4.

Set knobs 2 and 3 to the same position, 14 or more, according to the material used.

For plain knitting (jersey),

the knobs must be set especially well, or the stitch becomes very uneven.

Accidents during work. How to put them right.

This chapter is to be consulted whenever a mishap occurs during work.

1° Damaged needles.

Needle butts can be bent by the carriage:

a) if needles in the out-of-work position are not pushed down far enough,

b) if needles in the working position are not pushed up far enough,

c) if the needles have been put in work otherwise than by their springs,

Needles must never be put in work otherwise than by pushing them by their springs.

d) if safety springs are broken,

e) if pins 1 to 4 (fig. 2) are not fully in or fully out.

Needle hooks can be damaged:

a) by the passage of a large knot,

b) during the cast-on, if they catch on the steel wire of the comb.

Needle latches can be bent:

if the yarn-carrier tip is set too low and rubs on the needles,

if the brushes are badly set.

To replace a faulty needle,

take off the weights, remove the needle-retaining strip with the hook supplied for this purpose with the machine, and replace the needle in question with a new one, taking care to place the new needle high enough for the strip not to meet its latch. Put the stitch on the needle, replace the strip and the weights, and continue working.

To replace a safety spring,

unscrew the fixing screw of the spring cover strip nearest the faulty spring, then slip a screwdriver under the strip to lift it slightly. Take out the spring and insert a new one from below.

2° Broken yarn.

The yarn breaks (especially with yarns of inferior quality) in the following cases:

a) if it is badly wound (unevenly wound on the cone) or badly waxed,

b) if it is too tight or wrongly threaded through the tension unit,

c) if it has knots that are too large or badly made.

Join the two ends of the yarn with the weaver's knot or the reef knot (see fig. 14 and 15).

If a hole has appeared in the knitting, pick up the dropped stitches and, if necessary, undo a few rows (see page 19).

3° Badly formed stitches.

Stitches form badly or not at all:

a) if the knitting is too loose (knobs set too low) or too tight (knobs set too high),

b) if the yarn-carrier tip is badly set (too high or too low),

c) if the yarn carrier is twisted or bent and does not bring the yarn to the right place,

d) if the knitting is very wide and too little weight has been hung; the stitches are then faulty at the selvedges.
This fault is seen especially in tubular knitting; in this case one can also hook on to the selvedges the claws and small weights supplied with the machine.

e) if the latch of a needle opens with difficulty; the small groove of the needle must then be cleaned with a pin, or the needle replaced.

4° Stitches drop,

a) if one knits too fast, especially in full or half cardigan,
b) if the pull on the knitting is insufficient (add weights),
c) if the needles lack oil and jump in their grooves; this is what is called needle jumping.

5° The needles lift the knitting,

a) if the latch is bent; it can be straightened with pliers, or the needle changed,
b) if the knitting is too tight (knobs set too high),
c) if the weights are insufficient (see table page 14),
d) if one knits a harsh material without having waxed it.

6° The carriage jams,

a) if the butt of a needle out of work is not fully lowered,
b) if the carriage is brought back before the end of its pass, that is, before it has fully left the field of needles in work.

To free the carriage, find out which needle is jamming it by trying to move all the engaged needles one after another. For this operation it is better to remove the brushes, so as not to damage them.

Pull the jammed needle upwards or push it, while trying at the same time to move the carriage.

Sometimes, to free the carriage, there is nothing for it but to unscrew the guide bars and lift it. In that case, take care afterwards to put them back exactly in place.

If the carriage jams while working with the Super-8, see page 38.

7° The knitting is uneven,

a) if the knobs are not set at their exact position; for this setting to be correct, the eye must not look at an angle but be directly opposite the pointer *c*, fig. 7,
b) if the front bed is not in its normal position: pull or push the racking handle *F* fully home and pull the bed-lowering handle *H*, fig. 1;
c) if the yarn used is uneven (poor quality) or unevenly waxed;
d) if one knits at an uneven speed.

8° The knitting has vertical stripes,

a) if needles run stiffly, either because their butt has been bent by a knock or because their groove is clogged;
b) if needles are damaged (bent or stiff latch).

9° The selvedges are faulty,

a) if the tension is too weak or too strong;
b) if the yarn carrier is taken too far from the selvedges;
c) if the brushes are badly set and catch the yarn (see "Setting the brushes", fig. 8);
d) if the amount of weight hanging from the knitting is insufficient;
e) if the knitting is too tight (knobs set too high);
f) if the yarn carrier is bent, that is, twisted to the left or to the right. In the 1st case the selvedge will be faulty on the right; in the 2nd case it will be faulty on the left.

The setting of the yarn carrier is checked as follows:

Remove the rear brush.

Work the front needles only, and run the carriage with the yarn carrier *without yarn.*

Observe the position of the needle directly below the yarn-carrier tip: the tip of the latch must project at least 1 mm beyond the lower end of the yarn-carrier tip.

Make this check by stopping the carriage once in its pass from right to left, then once in its pass from left to right.

How to undo a few rows of knitting on the machine and how to pick up a dropped stitch.

Holes may sometimes appear in the knitting. If it is simply a dropped stitch, it can be picked up once the article is finished, with a machine needle, in the same way as for the chain cast-off. But as soon as a dropped stitch is noticed during work, take the precaution of securing it so that it runs no further; it is enough to pass a needle through the stitch to hold it.

If the damage is more serious and there is a real hole, caused for example by a large knot passing or by the yarn breaking, it is better to re-stitch the needles at the place where the hole occurred and then undo a few rows of knitting. The repair made in this way will be invisible, which would not be so if it were done afterwards.

To undo a few rows, first take off the weights hanging from the comb, then proceed as follows:

Take hold of the end of the yarn between the yarn-carrier tip and the needles and pull it gently in the direction in which the knitting comes undone, so as to raise the needles a little. At the same time the stitches of the last row but one will be brought one after another to the level of the needle heads (hook), and small jerks given to the yarn will pass these stitches into the hooks of the corresponding needles. With a little practice this can be done very quickly.

In the case of full cardigan knitting, before undoing, lower the needles in the half-drop position so as to form the stitches.

Ordinary 2 × 2 rib.

Set the racking handle *F*, fig. 1, in the middle.

Needle arrangement:

I . I I . I I . I I . I I . I I . I Rear
I I · I I · I I · I I · I I · I I Front

Take care that the outermost needles on the left and right are on the rear bed.

(A bar represents a needle in work, a dot a needle out of work.)

Push the lever *F*, fig. 1, to rack the front bed so as to obtain the following needle arrangement:

I . I I . I I . I I . I I . I I . I Rear
I I · I I · I I · I I · I I · I I Front

Cast on. (For 2 × 2 rib this must always be done with the needle arrangement above.)

Knit 2 or 3 tubular cycles to form the welt and stop the carriage on the left.

Rack the front bed back (return it to its normal position). Pull out pins 1 and 3 (fig. 2) and set the knobs as follows:

gauge 36 (No. 7)			gauge 32 (No. 8)		
knobs at	3 / 14	4 / 14	knobs at	3 / 13	4 / 13

Bring the carriage to the right and set the knobs as follows:

gauge 36 (No. 7)			gauge 32 (No. 8)		
knobs at	1 / 14	2 / 14	knobs at	1 / 13	2 / 13

The machine is now ready to produce ordinary 2 × 2 rib at the rate of one row of stitches, at each pass of the carriage, on all the needles in work (front and rear).

Because of its great elasticity, 2 × 2 rib is suited above all to borders

(ribbed borders of socks), cuffs and the bottom of pullovers, and in general to the parts of garments that must fit closely to the body.

Changing from 2 × 2 rib to knitting on one needle bed.

After finishing the 2 × 2 rib, put back in work the needles left out of work between the ribs on the rear bed.

With the transfer tool, transfer the stitches of the front bed to the rear bed, one on the empty needle, the other on its neighbour to the right.

Put the needles of the front bed out of work, or else close pins 1 and 4.

Set knobs 2 and 3 to 14 or more, according to the material used.

How to determine the number of needles and the number of carriage cycles for a given article.

As the exact figures for an article are not always to hand, proceed as follows:

Knit a small sample (in 1 × 1 rib, for example), 50 needles wide and 40 cycles high. Press this sample without stretching it, placing a damp cloth between the iron and the knitting.

On the piece of knitting obtained, count the number of stitches in a width of 10 cm. (for example 35 stitches). To obtain the total number of needles to put in work, multiply this number of stitches by the total width (for example 48 cm.) and divide by 10, that is:

$$\frac{35 \times 48}{10} = 168 \text{ needles}$$

In this case 168 needles must be put in work on the front and 168 on the rear to obtain a knitted piece 48 cm. wide in 1 × 1 rib.

Proceed in the same way to obtain the number of cycles.

Count on the sample the number of cycles in a height of 10 cm. (for example 30 cycles, which in 1 × 1 rib make 60 stitches). Multiply this number by the total height required, for example 62 cm., and divide by 10, that is:

$$\frac{30 \times 62}{10} = 186 \text{ cycles, to knit}$$

Pullover.

Lady's pullover, 1 × 1 rib with stripes (fig. 28).

Gauge 32. Wool 24/3. 380 grams. Medium size.

The pullover consists of:
one piece for the back, fig. 23,
one piece for the front, fig. 24,
two pieces for the sleeves, fig. 25,
two pieces for the neckband.

Back (fig. 23).

Set the knobs for 2 × 2 rib, that is:

$$\frac{3}{12} \times \frac{2}{12}$$

$$\frac{4}{12} \times \frac{1}{12}$$

Put 189 needles in work on the front and 189 on the rear.

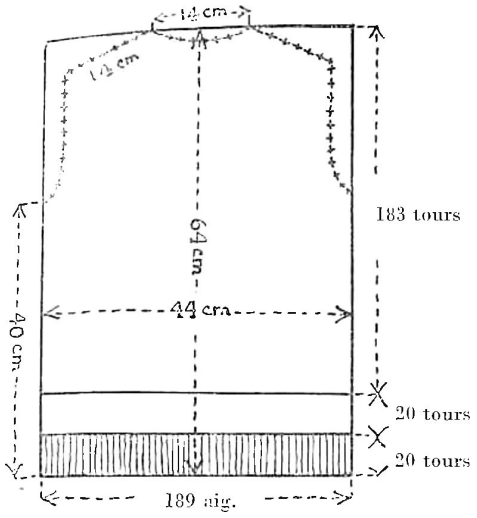


Fig. 23. Back.

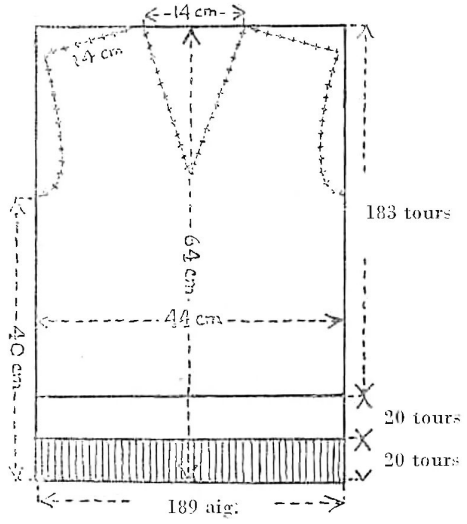


Fig. 24. Front.

Arrange them for 2 × 2 rib as follows:

I . I I . I I . I I . I I . I Rear
I I · I I · I I · I I · I I Front

Rack, cast on, knit 3 tubular cycles with pins 1 and 3 closed, then rack back, open pins 1 and 3 and knit 20 cycles of 2 × 2 rib.

Put back in work the needles left out of work between the ribs.

Close pins 1 and 3 and knit 1 cycle.

Open pins 1 and 3.

Set the knobs for 1 × 1 rib, that is:

$$\frac{3}{11\frac{1}{2}} \quad \frac{2}{11\frac{1}{2}} \times \frac{4}{11\frac{1}{2}} \quad \frac{1}{11\frac{1}{2}}$$

Knit ½ cycle, then make the stripes as follows:

3 cycles in wool of the other colour,
1 cycle " " " ground colour,
7 cycles " " " the other colour,
2 " " " " ground colour,
4 " " " " the other colour,
1 cycle " " " ground colour,
2 cycles " " " the other colour,
183 " " " " ground colour.

The piece is now finished.

Take the work off the machine and make a second identical piece for the front (fig. 24).

Sleeves (fig. 25).

Set the knobs for 2 × 2 rib, that is:

$$\frac{3}{12} \quad \frac{2}{12} \times \frac{4}{12} \quad \frac{1}{12}$$

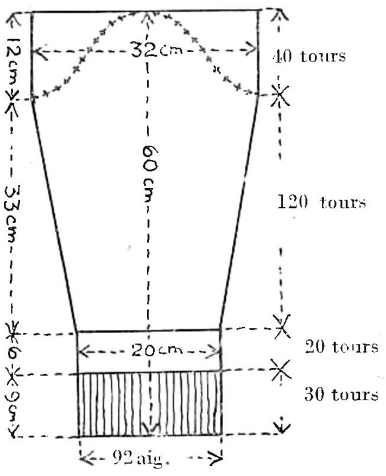


Fig. 25. Sleeves.

Put 92 needles in work on the front and 92 on the rear.

Arrange them for 2 × 2 rib as follows:

I . I I . I I . I I . I I . I I . I Rear
I I · I I · I I · I I · I I · I I Front

Rack, cast on, knit 3 tubular cycles with pins 1 and 3 closed, then rack back. Open pins 1 and 3 and knit 30 cycles of 2 × 2 rib.

Put back in work the needles left out of work between the ribs.

Close pins 1 and 3 and knit 1 cycle.

Open pins 1 and 3.

Set the knobs for 1 × 1 rib, that is:

$$\frac{3}{11\frac{1}{2}} \quad \frac{2}{11\frac{1}{2}}$$

$$\times$$

$$\frac{4}{11\frac{1}{2}} \quad \frac{1}{11\frac{1}{2}}$$

Knit ½ cycle, then make the stripes as follows:

3 cycles in wool of the other colour,
1 cycle " " " ground colour,
7 cycles " " " the other colour,
2 " " " " ground colour,
4 " " " " the other colour,
1 cycle " " " ground colour,
2 cycles " " " the other colour,
½ cycle " " " ground colour.

Increase on each side every 4 cycles, 30 times, as follows:

Put one more needle in work, on the front and on the rear in turn, *on the carriage side*, that is:

Carriage on the left. Increase one needle left rear,

Carriage on the right. Increase one needle right front,

Carriage on the left. Increase one needle left front,

Carriage on the right. Increase one needle right rear.

Knit 2 more cycles and repeat in this way 30 times, until there are 152 needles and 120 cycles.

Knit 40 more cycles straight.

The piece is now finished.

Take the work off the machine and make a second identical piece for the second sleeve.

Neckband:

Set the knobs for 2 × 2 rib, that is:

$$\frac{3}{12} \quad \frac{2}{12}$$

$$\times$$

$$\frac{4}{12} \quad \frac{1}{12}$$

Put 189 needles in work on the front and 189 on the rear.

Arrange them for 2 × 2 rib as follows:

I . I I . I I . I I . I I . I I . I Rear
I I · I I · I I · I I · I I · I I Front

Rack, cast on, knit 3 tubular cycles with pins 1 and 3 closed, then rack back.

Open pins 1 and 3 and knit 10 cycles of 2 × 2 rib.

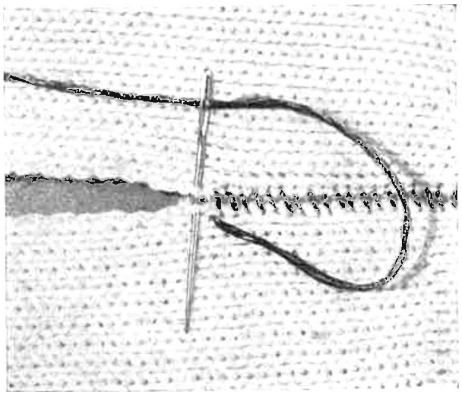


Fig. 26. Overcasting.

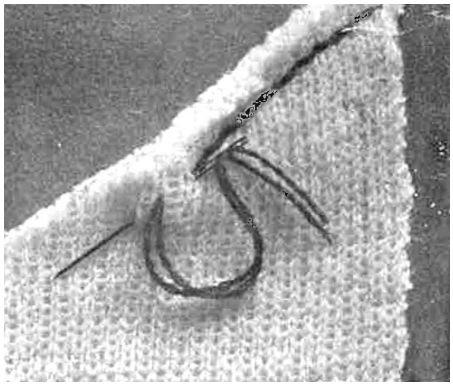


Fig. 27. Backstitch.



Fig. 28. Finished pullover.

Close pins 2 and 3, knit 2 cycles in this way on the front, open pins 2 and 3, and knit 10 more cycles.

Take the work off the machine and make a second identical piece.

Making up (fig. 26, 27 and 28).

(See also page 43.)

All the pieces must be pressed along the knitting with a hot iron, placing a damp cloth between the knitted piece and the iron.

Cut the various pieces according to the diagrams.

Join the front and the back with overcasting stitches (fig. 26) at the sides and with backstitch (fig. 27) at the shoulders.

Close the sleeves likewise with overcasting stitches and set them in with backstitch. Cover all seams without selvedges with seam tape of the same colour as the wool, pressing them open flat.

For the neckband, join the two pieces end to end to form a single long strip, then join the two ends into a point sewn with backstitch.

Fit the neckband with the cast-on edge on the outside and sew it on with side stitches.

Table of measurements for ladies' pullovers.

Size	Height		Width		Sleeve		
	Back	Front	Back	Front	Height	Bottom width	Top width
42	58	58	44	45	59	19	32
44	61	61	45	47	60	21	34
46	65	65	46	49	61	23	36

Table of measurements for ladies' cardigans.

Size	Height		Width		Sleeve		
	Back	Front	Back	Front	Height	Bottom width	Top width
42	60	62	44	23	59	20	33
44	63	65	46	24	60	22	35
46	66	68	48	26	61	24	37

Pleats.

Pleated skirts, so pretty but so delicate in woven cloth, can be knitted with the greatest ease. They then have the advantage of needing no pressing with an iron, the pleats being formed by the knitting itself, and of keeping their pleats even after washing. There are different kinds of pleats, shown in the examples below:

Knife pleats.

Cast on all the needles and knit 2 cycles of welt. With the transfer tool (see page 30), transfer the stitches of the needles shown by a dot in the diagram below to the neighbouring needles, and lower the free needles.

The needles will be arranged as follows:

I . . I I I I I I I I . . I I I I I I I I . . I I I I I I I I
I I I I I I I I · · I I I I I I I I · · I I I I I I I I · · I etc.

Pull out pins 1 and 3.

Set the knobs as for 1 × 1 rib and leave them free.

The machine is ready to produce "*knife-pleated knitting*".

Box pleats.

Needle arrangement:

* * * *
I I I I I I I I I I . I I I . I I I I I I I I I I . I I I . I I
I I I · I I I · I I I I I I I I I I · I I I · I I I I I I I I I I etc.
* * * *

Cast on and place the comb.

Remove the yarn from the needles marked with a *.

Knit one cycle, pulling the comb with the left hand.

Hang the weights.

Knit the welt (2 or 3 cycles).

Set the knobs as for 1 × 1 rib and leave them free.

The machine is ready to produce "*box-pleated knitting*".

Separating yarn.

To make a pleated skirt of medium size, 4 pieces must be joined.

The 4 pieces of the skirt are made joined to one another, using the separating yarn.

After finishing the first piece, put back in work all the needles left out of work for the pleats.

Bring the carriage to the right and close pins 1 and 3.

Change yarn carrier. (Silk or cotton will have been threaded in the second yarn carrier beforehand.)

Knit one tubular cycle, open pins 1 and 3 and change yarn carrier.

Knit 1 row.

Close pins 1 and 3. Knit 2 tubular cycles.

Remove needles for the pleats with the transfer tool, as before.

In this way there is a perfect cast-on for each pleated piece.

This separating yarn can be used for any pieces of knitting.

Sleeved jacket (fig. 29).

Gauge 32. White wool 2/24. Weight 230 grams.



Fig. 29. Jacket.

The jacket consists of:
One piece for the body, fig. 30, and two pieces for the sleeves, fig. 31.

Body (fig. 30).

Put 128 needles in work on the front and 128 on the rear.

Cast on and knit 5 tubular cycles.

Set the knobs for 1 × 1 rib.

On the front bed, transfer to the neighbouring needles the stitches of the 21st and 25th needles, counted from the selvedge on each side. Put the empty needles out of work.

On the rear bed, transfer to the neighbouring needles the stitches of the 2 needles facing the groups of 3 needles between the needles lowered on the front. Put the empty needles out of work.

Knit 180 cycles.

On the front bed, transfer the stitch of needle No. 1 on the right to needle No. 1 on the left (middle needles), without putting the empty needle out of work.

Knit one cycle.

Transfer the stitches of needles No. 3 to needles No. 2 and the stitch of needle No. 1 on the right to needle No. 1 on the left, still without putting the empty needles out of work.

Knit one cycle.

Transfer the stitches of needles 5, 3 and 1 R (right) to needles 4, 2 and 1 L (left).

Knit one cycle.

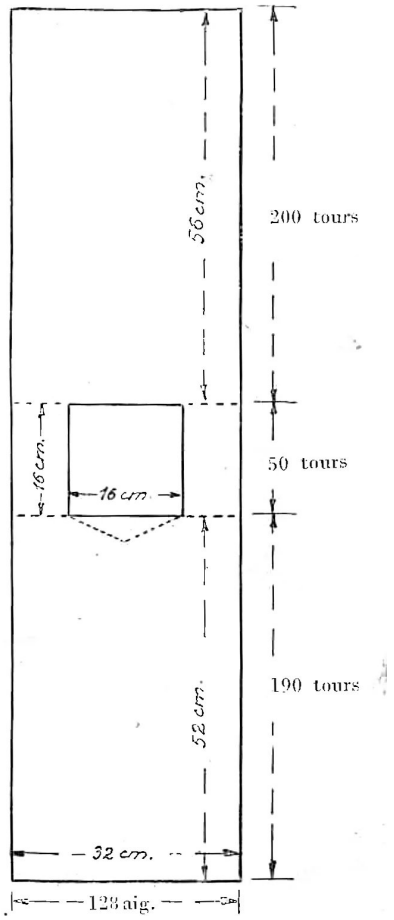


Fig. 30. Body of the jacket.

Transfer the stitches of needles 7, 5, 3 and 1 R to needles 6, 4, 2 and 1 L.

Knit one cycle.

Continue in this way up to needle No. 20, never putting the empty needles out of work.

Knit one cycle.

Make the chain cast-off on 50 stitches at the centre (see page 27).

Put the empty needles out of work.

Knit 50 cycles on the left only.

Knit 50 cycles on the right only.

Put back in work the 50 needles at the centre previously put out of work, cast on again and put the comb back.

Knit 200 cycles and finish with the chain cast-off.

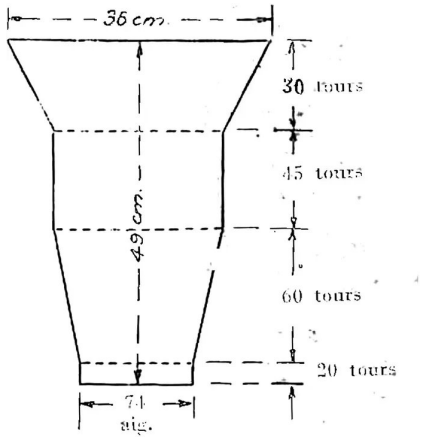


Fig. 31. Sleeve.

Sleeves (fig. 31).

Put 74 needles in work on the front and 74 on the rear.

Cast on and knit 5 tubular cycles.

Set the knobs for 1 × 1 rib.

Knit 20 cycles.

Increase 15 times, every 4 cycles, on the left and on the right.

Knit 45 cycles.

Increase 12 times, every 2 cycles, on the left and on the right.

Knit 6 cycles and finish with the chain cast-off.

Making up.

Join the two parts of the jacket with overcasting stitches, close the sleeves and set them in.

Edge the neck opening and the bottom of the sleeves with a small crochet picot.

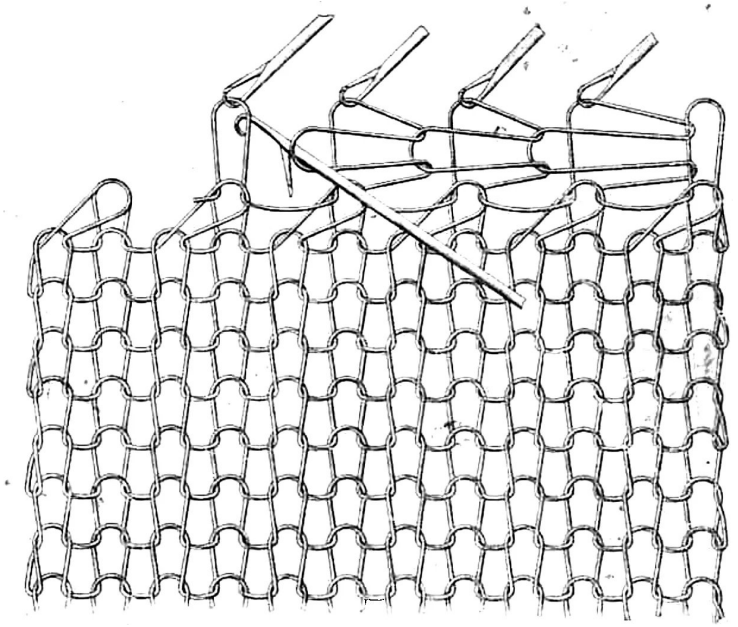


Fig. 32. Chain cast-off.

Chain cast-off done entirely on the machine (fig. 32).

This arises mainly when an opening must be made for a neck, an armhole, etc., or in general when a certain number of needles must be put out of work while knitting an article.

Proceed then as for the first method (page 15), that is:

on the part where the chain cast-off is to be made, transfer the stitches of the front needles to the corresponding rear needles,

knit one row with all 4 pins pulled out,

release the loops formed by the front needles on the part where the chain cast-off will be,

lower the front bed.

With a machine needle, form the chain cast-off as shown in fig. 32, that is, pass the needle through the first, then through the second stitch from the right, leaving the tip of the latch between the two stitches; withdraw the needle, hooking the right strand of the second stitch; pass this second stitch through the first.

Continue in the same way, passing the third stitch through the second, the fourth through the third and so on, taking on the needle only the right strand of each stitch.

On reaching the last stitch, pass the end of the yarn through it (see fig. 22).

If the chain cast-off is made on the right-hand part of an article, transfer the last stitch to the next needle.

Finally release the stitches of the needles where the chain cast-off was made. We recommend a few practice runs before making the chain cast-off on fashioned articles.

Full cardigan.

1 × 1 full cardigan or double rib.

Same needle arrangement as for ordinary 1 × 1 rib. Cast on, knit the welt and stop the carriage on the left.

Pull out pins 1 and 3, fig. 2, and set the knobs as follows:

gauge 36 (No. 7)			gauge 32 (No. 8)		
knobs	3	4	knobs	3	4
at	16	3	at	14	3

Bring the carriage to the right and set the knobs as follows:

gauge 36 (No. 7)			gauge 32 (No. 8)		
knobs	1	2	knobs	1	2
at	16	3	at	14	3

Work the carriage evenly: at each cycle (one outward pass and one return pass) it produces, simultaneously, a row of double stitches on the front and a row of double stitches on the rear.

Half cardigan.

1 × 1 half cardigan or pearl rib.

Same needle arrangement as for ordinary 1 × 1 rib. Cast on, knit the welt and stop the carriage on the left.

Pull out pins 1 and 3, fig. 2, and set the knobs as follows:

gauge 36 (No. 7)			gauge 32 (No. 8)		
knobs	3	4	knobs	3	4
at	13	13	at	12	12

Bring the carriage to the right and set the knobs as follows:

gauge 36 (No. 7)			gauge 32 (No. 8)		
knobs	1	2	knobs	1	2
at	16	3	at	14	3

Work the carriage evenly: at each cycle (one outward pass and one return pass) it produces, simultaneously, 2 rows of single stitches on the front and a row of double stitches on the rear.

Racked patterns.

By combining the needle arrangement with the racking of the bed, a pretty variety of fancy patterns can be produced in full and half cardigan.

Note that the knob positions will be looser for racked knitting than for ordinary knitting. In addition, knobs No. 1 and 3 are **always left free** (see page 7) in full cardigan, and knob No. 1 in half cardigan.

Racked patterns in full cardigan.

1st example:

Cast on and knit the welt as usual, making sure that, with the racking handle *F*, fig. 1, in the middle, the needle arrangement is indeed the one given under "Casting on", page 12.

Set the knobs as for full cardigan, but knob No. 1 one division lower, for example:

gauge 36 (No. 7)		gauge 32 (No. 8)	
3	2	3	2
16	4	15	4
×		×	
4	1	4	1
4	17	4	16

Rack at each cycle, when the carriage is on the left, once by pushing the lever *F*, fig. 1, once by pulling it. Always work the handle fully home.

2nd example:

Proceed as above, but rack only every 2 cycles, instead of every cycle.

3rd example:

Arrange the needles, cast on and knit the welt as given in example 1. Set the knobs as follows:

gauge 36 (No. 7)		gauge 32 (No. 8)	
3	2	3	2
17	4	16	4
×		×	
4	1	4	1
4	17	4	16

Rack at each row, eleven times in succession.

Knit one row without racking.

Rack at each row, eleven times in succession.

Knit one row without racking and so on.

This gives a zigzag whose size can be varied by changing the number of racked rows between two unracked rows.

4th example:

Cast on in 2 × 2 rib, as shown on page 16, then set the knobs as in example 1.

Rack at each cycle or every two cycles, *when the carriage is on the left.*

5th example:

See that the racking handle is pulled forward. Arrange the needles as follows:

I I I . I I I . I I I . I I I Rear
I I · I I I · I I I · I I I · I Front
etc.

Cast on; after placing the comb, remove the yarn from the middle needle of each group of 3 needles, front and rear.

Knit the welt, then set the knobs as for full cardigan (see example 1).

Rack at each cycle, *when the carriage is on the left.*

Note:

For a stronger edge, cast on and knit the welt on all the needles, then with the transfer tool transfer the stitches of the needles to be lowered to the neighbouring needles.

6th example:

See that the racking handle is pulled forward. Arrange the needles as follows:

IIIIIII..IIIIIII..IIIIIII Rear
I·IIII·II·IIII·II·IIII· Front
etc.

Begin as shown in the note above.

Set the knobs as for example 1.

Rack at each cycle, *when the carriage is on the left.*

7th example:

Cast on and knit the welt, and set the knobs for half cardigan. With the transfer tool, arrange the needles as follows:

I . I . I . I . I . I . I . I . I . I . I . I . I Rear
III · I · I · I · II · I · I · I · II · I · etc. Front

The outermost needle on the left will be on the front bed and the outermost needle on the right on the rear bed.

Knit 8 cycles (16 rows), racking one notch after each cycle. Rack once to the right and once to the left.

Knit 1 cycle (9th cycle). Rack one notch in the same direction as for the 8th cycle. (You will thus have racked 2 times in the same direction.)

Knit 7 cycles, again racking after each cycle, once in one direction and once in the other.

Knit 1 cycle (17th cycle). Rack in the same direction as for the 16th cycle, and so on.

In this way a very pretty checkerboard effect is obtained in a racked pattern.

8th example:

Cast on and knit the welt as usual.

Set the knobs as for very loose ordinary 1 × 1 rib.

With the transfer tool, arrange the needles as follows:

I I I I I I I I I I I I I I I I I I I I Rear
I I I I · · · · I I I I · · · · I I I I etc. Front
1 2 3 4 5 6 7 8

The stitches of the lowered needles 1, 4, 5 and 8 etc. are transferred to the neighbouring needles on the left and right respectively, while the stitches of needles 2, 3, 6 and 7 are transferred to the opposite needles of the rear bed.

Place the bed fully to the right, that is, with the racking at the last notch to the right.

Knit 3 cycles (6 rows) of ordinary 1 × 1 rib.

Push in pins 2 and 3.

Knit 4 rows, racking one notch to the left after each row.

Pull out pins 2 and 3.

Knit 3 cycles of ordinary 1 × 1 rib.

Push in pins 2 and 3.

Knit 3 rows, racking one notch to the right after each row, and so on.

9th example:

Place the front bed fully to the left by pulling the racking lever right out.

Put in work, on the front and on the rear, a number of needles divisible by 7.

Cast on and knit the welt. Carriage on the left.

Set the knobs as follows:

gauge 36 (No. 7)			gauge 32 (No. 8)		
Knobs	3	4	Knobs	3	4
at	13	13	at	12	12

Bring the carriage to the right and set the knobs as follows:

gauge 36 (No. 7)			gauge 32 (No. 8)		
Knobs	1	2	Knobs	1	2
at	17	4	at	16	4

Knob No. 1 free.

With the transfer tool, arrange the needles as follows:

I I I I I I . I I I I I I . I I I I I I . Rear
I I I I I I · I I I I I I · I I I I I I · etc. Front

A group of 7 needles remains on the left.

Knit 3 cycles, increasing by one needle *on the left front* each time the carriage is on the left.

On the left rear, transfer the stitch of the selvedge needle to the neighbouring needle and put the free needle out of work.

The needles will then be arranged as follows:

I I I I I I . I I I I I I . I I I I I I . Rear
I I I I I I I I I · I I I I I I · I I I I I I · etc. Front

Knit 4 cycles without racking, 4 cycles racking one notch to the right at each cycle — 4 cycles racking one notch to the left at each cycle — 4 cycles without racking, and so on.

Variant:

Knit 4 cycles without racking, — 4 cycles racking one notch at each cycle to

the right — 4 cycles without racking — 4 cycles racking one notch at each cycle to the left, and so on.

Racked patterns in half cardigan.

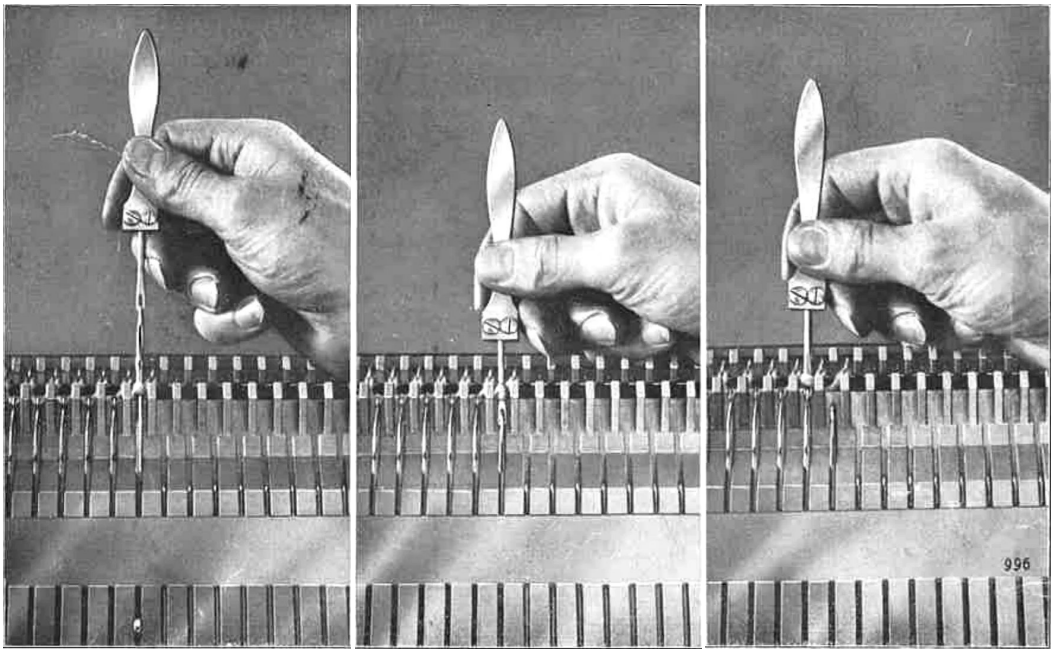
All patterns racked at each cycle can be made in half cardigan as well as in full cardigan (see examples 1, 2, 4, 5, 6, 7 and 9).

Notes.

Taking inspiration from the foregoing examples, all sorts of other patterns can be created, either by choosing other needle arrangements or by varying the racking.

Racked knitting is used for fancy articles such as children's suits, men's waistcoats, etc.

Increasing and decreasing.



Using the transfer tool.

Fig. 33. Fig. 34. Fig. 35.

Using the transfer tool.

To obtain directly on the machine the proper shape of certain articles or parts of articles, increases or decreases are made, using the working hook and the decreasing transfer tool.

The use of the latter is shown in detail in fig. 33, 34 and 35.

Hold the tool in the right hand and, with the left hand, push up by their butts the needles on which the transfer is made, so as to open the latches and free the hooks.

Fig. 33. Catch the needle by its hook with the tool and pull it up until the stitch is behind the latch.

Fig. 34. Bring the needle down, pushing it with the tool, until the stitch has slid onto the tool. Free the tool from the needle hook without letting go of the stitch.

Fig. 35. Lift the tool and, keeping firm hold of the stitch, carry it to the neighbouring needle.

Tip the tool so that the stitch drops into the needle hook.

The transfer tool is used for all kinds of work: decreases, stitch holding, openwork and plated patterns, chain cast-off, pleats, etc.

Increases.

Increases can be made by either of the following two methods:

First method. Put one more needle in work, on the front and on the rear in turn, *on the side where the carriage is.*

Increase for example as follows:

Carriage at left: one needle left rear.
" " right: " " right front.
" " left: " " left front.
" " right: " " right rear.

Space the increases according to the shape to be obtained, that is, increase every 2, 3, 4 or more cycles.

This method is the quickest, but it does not give very fine selvedges.

Second method. Put one more needle in work on the front and on the rear. With the transfer tool, transfer to each of these needles the stitch of the neighbouring needle. Bring back to the needles thus freed the last stitch but one that they had formed, picking it up with the working hook.

Increases made by this method take longer, but they give more even selvedges.

This method also allows increases on both sides at once, whereas with the previous method one can increase only on the side where the carriage is.

Decreases.

For decreases, the two- or three-point transfer tools are preferably used, so that the double stitches resulting from the transfer lie a little inside the selvedge.

The work is similar to that of the simple one-point tool shown in figures 33 to 35, that is:

- with the left hand, raise the 3 or 4 selvedge needles a little to open their latches,
- catch the 2 or 3 selvedge needles with the tool, pull them up and then bring them down so as to pass the stitches onto the tool,
- transfer the stitches one needle further on, towards the middle of the knitting, pass them into the needle hooks and lower by its butt the needle that has become free.

To avoid being hindered by the carriage, decrease preferably on the right when the carriage is on the left, and on the left when the carriage is on the right. One can also decrease on both sides at the same time.

Transfer patterns.

Example of a transfer pattern.

Our diagram shows the initial position and the successive transpositions of the needles of the front bed. The small black squares represent the needles in work, the white squares the needles out of work.

The instructions given concern only the pattern proper, which must of course be preceded by a cast-on and possibly by a plain or 1 × 1 rib border.

For brevity, the expressions "Push the pin in" and "Pull the pin out" are replaced by the simpler "Close the pin" and "Open the pin".

Set the knobs for ordinary 1 × 1 rib.

Rear bed: all needles in work.

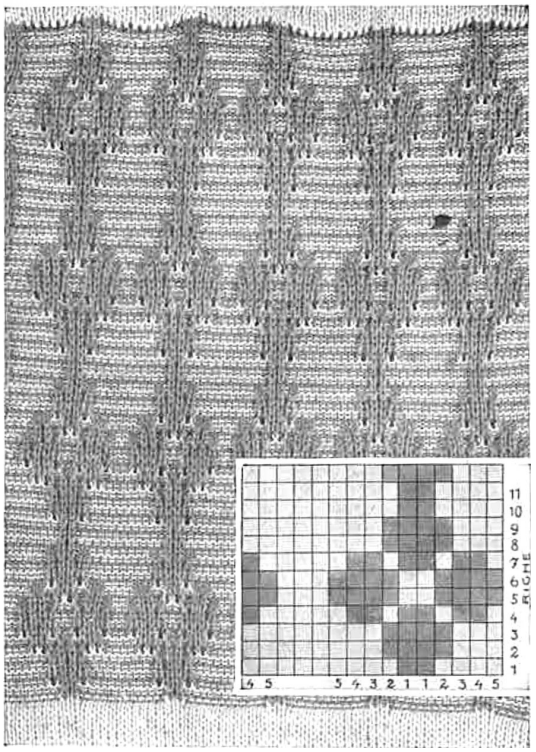


Fig. 36.

Front bed: 2 needles in work, 11 out of work, and so on.

Knit one cycle with all pins open. Change yarn carrier.

Close pins 1 and 4.

Knit 1 cycle, change yarn carrier (1st row of the pattern).

Put needles 2 in work (see diagram).

* { Open pins 1 and 4, knit one cycle, change yarn carrier.
Close pins 1 and 4, knit one cycle, change yarn carrier (2nd row).

Repeat the cycles marked with the sign * (3rd row).

Transfer the stitches of needles 2 to the corresponding needles of the rear bed. Put the empty needles out of work.

Put needles 3 and 4 in work.

Knit the cycles marked with the sign * (4th row).

Transfer the stitches of needles 1 to the corresponding needles of the rear bed. Put the empty needles out of work.

Put needles 2 and 5 in work.

Knit 2 times the cycles marked with the sign * (5th and 6th rows).

Transfer the stitches of needles 2 and 5 to the corresponding needles of the rear bed.

Put the empty needles out of work.

Put needles 1 in work.

Knit the cycles marked with the sign * (7th row).

Put needles 2 in work.

Knit 2 times the cycles marked with the sign * (8th and 9th rows).

Transfer the stitches of needles 2 to the corresponding needles of the rear bed. Put the empty needles out of work.

Knit 2 times the cycles marked with the sign * (10th and 11th rows).

The motif is finished. Repeat it as many times as desired, each time starting the operations again from the beginning.

Plating.

So-called "plated" knitting is made of two yarns of different kinds, for example a silk yarn and a cotton yarn, or a wool yarn and a cotton yarn, or else two yarns of different colours.

These two yarns are knitted simultaneously in such a way that one covers the other.

In tubular knitting the first will be visible on the right side, while the other will be visible on the wrong side.

In ribbed knitting (1 × 1, full cardigan, etc.) one of the yarns will be visible on both faces of the knitting, while the other will be hidden inside.

To obtain plated knitting, a special yarn-carrier tip called a *plating tip* is used, which simply replaces the ordinary tip (see fig. 37).

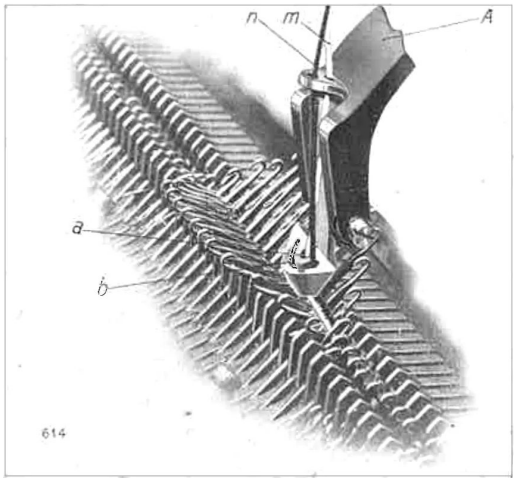


Fig. 37. Plating yarn-carrier tip.

The plating tip is used for example for knitting articles in half-wool (wool m, cotton n, fig. 37), or to reinforce the toe and heel of stockings and socks (the reinforcing yarn is passed through the half-moon opening), or else to obtain colour effects (transfer patterns).

Plating is also useful when an article is knitted from two different materials of which only one is to be visible, such as a silk and cotton article meant to look like pure silk, or ladies' winter stockings in wool and silk, which are to look like silk stockings while being warmer than ordinary silk stockings.

The yarn that is to be visible on the knitting (yarn *m*) is passed through the central hole a of the tip; the other (yarn *n*), which is to be hidden, is passed through the half-moon opening. Each of the two yarns must have its own tension, so two tension units are needed for each plating tip.

To obtain good plating, the following precautions must be taken:

1° check that the tip is exactly on the centre line of the machine, 1 mm. above the needle latches (see fig. 38);

2° give more tension to yarn *n*, in the half-moon entry, than to yarn *m*;

3° wax the materials carefully so that they do not risk tangling;

4° hang more weight from the comb than for ordinary knitting.

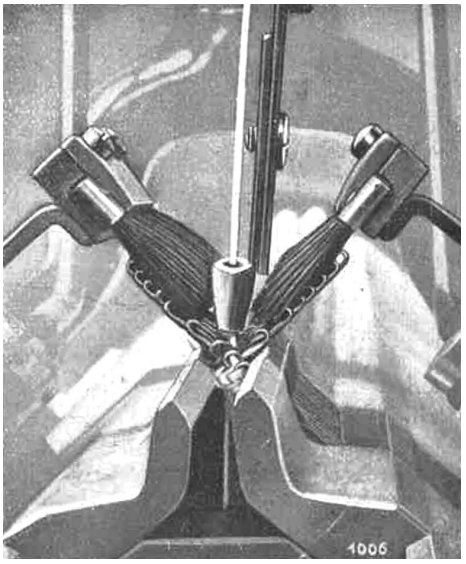


Fig. 38. Setting the yarn-carrier tip.

The plating tip is not included in the normal accessories; it is supplied on request, as is the additional tension unit.

This tip, generally little known, is of great service and we recommend it to every knitter, for it allows her to increase at little cost the variety of articles produced on the machine.

The 6-colour push-button yarn changer.

The MM, MR and CS machines normally work in 2 colours, but nothing is simpler than to raise this number to 6. One need only fit to the machine a small auxiliary device called the **six-colour** PUSH-BUTTON YARN CHANGER.

This device is fixed on the front bar, in place of the usual yarn carrier. The rear yarn carrier is removed and replaced by a special yarn carrier fitted with an open tip, which can take up or

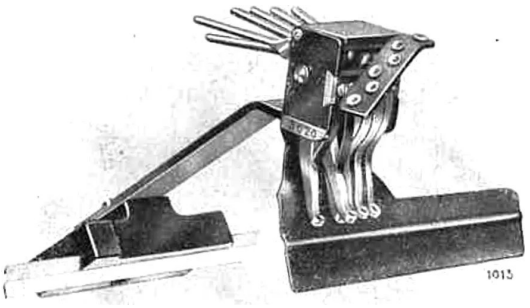


Fig. 39. Push-button yarn changer.

drop, as the knitter wishes, the various yarns of the push-button yarn changer, according to which button is put in work.

To take a colour, simply press the corresponding button when the carriage is on the right. This movement automatically puts out of work the yarn that was working before.

With each push-button yarn changer 4 additional tension units are supplied, since the standard machine has only 2 tension units.

Using the push-button yarn changer.

Place the push-button yarn changer on the front bar, at the right-hand side of the machine, so that the protective plate below the yarn-carrier tips lies between the two needle beds. The left end of this plate must be 2 or 3 mm from the selvedge needle.

Place on the rear bar the yarn carrier fitted with the open tip. Check its setting: for this, push up to the top of the bed one front needle and one rear needle, and bring the tip above these two needles.

It must be exactly in the middle, about 1 mm. from each needle, and must on no account touch the latches when open and folded back against the needles.

Check also the position of the push-button yarn changer: a yarn-carrier tip of the changer, in working position, must be exactly above the centre line of the machine (see fig. 40).

The underside of the tips out of work must be at about the same height as the top of the open tip.

Place additional tension units on the tension mast so as to have one tension unit per yarn. This arrangement is shown in fig. 1.

Pass the yarns coming from the tension units through the push-button yarn changer so that there is no contact between them.

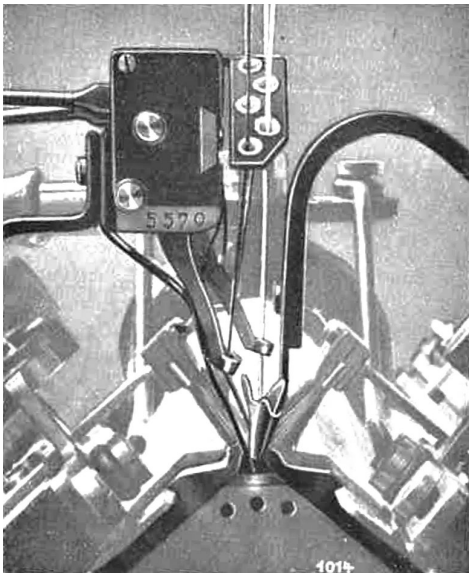


Fig. 40. Setting the push-button yarn changer.

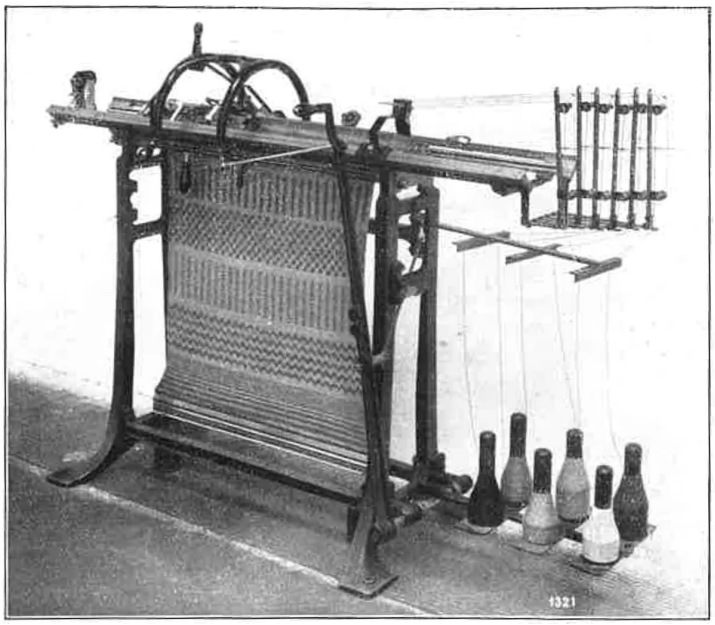


Fig. 41. CS machine with push-button yarn changer and Super-8.

Push the yarn changer handle to the left so as to put the special rear yarn carrier in work; from then on, this handle must not be moved. The colour is changed simply by pressing, when the carriage is on the right, the corresponding button of the push-button yarn changer.

Note on the CS machine.

The 6 tension units are generally placed to the right of this machine. They are carried by a kind of bracket fixed by two small blocks at the end of the carriage guide bars. This assembly is shown very clearly in fig. 41 and is easily understood without further explanation.

THE SUPER-8.

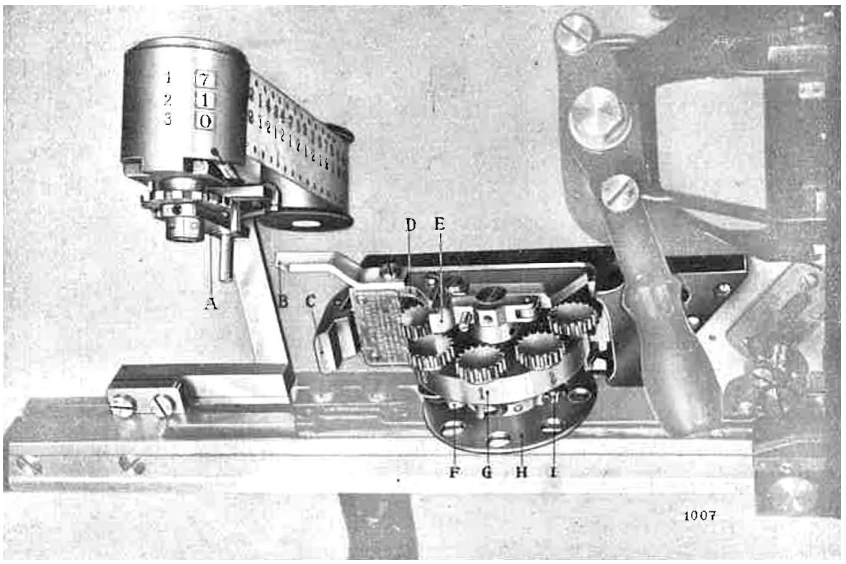


Fig. 42. Super-8.

The Dubied SUPER-8 is a new patented device that can be fitted to all Dubied hand machines of types MM, MR and CS in gauges 25, 28, 32 and 36. It makes it possible to produce on these very simple machines, without special training, patterned knitting with the same appearance as Jacquard knitting.

The Super-8.

The SUPER-8 consists of a support or frame in which the body *D*, fig. 42, pivots. The device shown in this figure is in its "in work" position, since the *body D* is engaged in the *hook C*. To put the device in the "out of work" position, the hook *C* is pushed to the left and the body *D* is lifted by two springs.

The essential part of the SUPER-8 is the *drum G*, bearing the *selector cams I*. The *lever H* is used to turn the drum by hand; just pull it towards you, then turn it in one direction or the other so as to bring the required number to the top of the drum. In fig. 42, the drum is in position 1.

The *selector cams* are small rotary cams of extremely precise manufacture. Their edge, cut to suit the needle selection to be obtained, acts on the needle butts and gives them an impulse either towards the top or towards the bottom of the bed.

There are 8 of these selector cams, whereas the old device called the Dubied Super-Auto-Rayeur had only 4. The variety of patterns produced by the SUPER-8 is thus much greater.

The SUPER-8 is supplied with two sets of selector cams: a set of 8 selector cams already mounted on the drum, which makes it possible to create a large number of motifs, and set *0*, consisting of two inverted selector cams used for making checks and vertical stripes.

To knit checks or vertical stripes, remove the two selector cams in positions 5 and 6 and replace them with the two inverted selector cams of set *0*. (See pages 39 and 40: "Changing selector cams.")

In addition to these two sets, there are several further sets of 8 selector cams. Each of these sets gives new series of patterns, so that knitters have every interest in owning several

sets, thereby increasing the output of their machine. These further sets can be obtained already mounted on complete drums: to work with another set, one then need only change the drum. For this, pull the two levers *E* and *F*, fig. 42, and lift the bearing covers.

Those who wish to save the cost of a spare drum can buy the selector cams alone and fit them themselves on the drum of their machine in place of the existing cams; this naturally takes more time (see pages 39 and 40, "Changing selector cams").

The pattern indicator

fixed to the front bar, in the path of the SUPER-8, consists of an endless *card strip* driven by a *toothed wheel*. At each carriage cycle the *ramp B*, fig. 42, works the *lever A* and moves the strip on one notch. One, two or three series of figures are printed on the strip, and they appear in turn in the three *windows* provided for this.

The figures appearing in *window 1* show *the order of the selector cams* to be put in work to obtain a given pattern.

The figures in *window 2* relate to the *yarn changer* and give the numbers of the colours to be used.

Finally, the figures in *window 3* relate to the SUPER-8 *racking attachment*, a device less often used (see page 62).

For most 2-colour patterns, the change of yarn carrier is made regularly at each carriage cycle. Window 2 is then not needed.

Those who do not use the racking attachment may if they wish use window 3 for marks relating to increases or decreases.

With each set of selector cams, the Dubied firm supplies a number of card strips corresponding to some of the motifs produced by that set. Those who wish to create new motifs themselves can obtain blank strips and write on them the sequence of selector cams for special patterns.

Preparing the Super-8

Before beginning work, check that the drum *G* carries the selector cams corresponding to the sample to be reproduced. Each selector cam bears a number stamped on one of its faces. On the back of the indicator paper strips is printed a table showing for each gauge the location of the selector cams on the drum. At the bottom of the table is shown the selection made by each cam.

If the selector cams must be changed, refer to the explanations on pages 39 and 40.

Cast on, knit the welt and stop the carriage on the left. With the required selector cams on the drum, check their position by trying to turn them on their axis; if the resistance is marked, they are correctly placed; if not, turn them until resistance is felt.

Put the SUPER-8 in work by pressing on the body *D*, fig. 42.

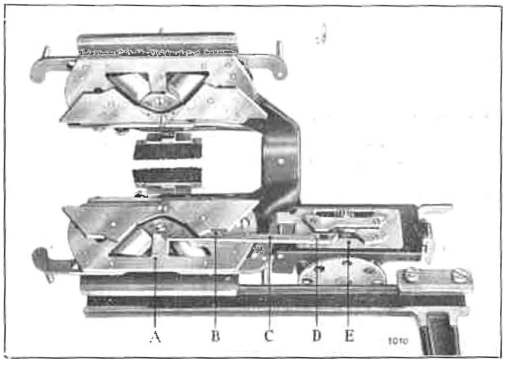


Fig. 43.
Dubied SUPER-8 seen from below.

Push pin 4 in.

Move knob No. 4 (left front) as high as possible.

Fit the connecting cam *C*, fig. 43, by first placing it on the headless screw *D*, then pressing it to the left until the small plate *B* enters the lower part of the guide entry of stitch cam No. 4.

Make sure that the plate *B* has entered fully into the entry. While holding the cam *C* with the left hand, unscrew knob No. 4 and let it descend; it will come to

rest automatically on division 4. Tighten knob No. 4.

On a type MM machine, the fitting of the connecting cam is now complete. On an MR or CS machine, knob No. 6 must also be placed as high as possible and tightened so that it cannot descend during work.

Set the knobs as follows:

gauge 36 (No. 7)		gauge 32 (No. 8)	
3/16	2/13	3/15	2/12
×		×	
4/4	1/13	4/4	1/12

Knobs No. 1, 2 and 3 must be free.

The preparation of the machine and the SUPER-8 is complete.

Place on the indicator the strip corresponding to the motif to be obtained, making sure that the number printed inside the strip is that of the chosen motif. To fit the strip, one need only lift the indicator lid (fig. 42). Place in the strip the small wooden roller that keeps it taut. Free the wheel by lifting the horizontal lever behind lever *A* and turn the wheel by hand to bring the strip to its starting point.

To place the indicator in its exact position, one need only push it along the bar until it is stopped:

on MM or MR machines, by the carriage placed fully to the left;

on CS machines, by its pin.

Knitting with the Super-8

Set the drum to the position shown by the first number on the indicator strip (window 1).

Knit one row from left to right, bring the device fully clear of the needles on the right, and change yarn carrier by working the yarn changer lever on the carriage bridge.

If working with the push-button yarn changer, change yarn by pressing the button of the yarn wanted. The number of this button is shown in window 2.

Knit one row from right to left, bringing the carriage far enough to the left for the device to be clear of the bed and for the indicator to be worked.

Turn the drum to bring to the top the number shown in window 1, knit one row from left to right, and so on.

Vertical stripes or checks.

To obtain vertical stripes or checks, use two similar selector cams (set 0), one of which makes the reverse selection of the other.

One need only fit them at locations 5 and 6 of the drum and place on the indicator the strip corresponding to the motif to be obtained. Motif No. 21760 is obtained by always working with the same selector cam and changing colour at each cycle.

Important notes

Putting the Super-8 in work.

The SUPER-8 cannot be put in work when the carriage is fully to the left, because the ramp *B* fouls the lever *A* (fig. 42). The carriage must therefore be brought a little to the right before putting the SUPER-8 in work.

Changing colour.

The colour is changed regularly at each cycle, when the carriage is on the right, by changing the position of the yarn changer lever or by pressing the buttons of the push-button yarn changer, whose numbers are shown in window 2 of the indicator.

Carriage jammed.

If the carriage is jammed by a needle during the pass from left to right, put the device out of action, bring the carriage to the right and finish the interrupted selection by hand. Change the needle that caused the jam, if necessary.

Bring the carriage to the left, put the device back in work and turn the drum to the next position shown on the card strip. Return to its stop notch the selector cam used for the previous row.

Running the carriage empty.

Never run empty, that is, without taking a yarn carrier along with the carriage, when the SUPER-8 is in the working position.

Never knit with the SUPER-8 in work without having fitted the connecting cam.

Changing from Super-8 knitting to ordinary knitting

Stop the carriage on the left.

Take hold of the connecting cam with the left hand, without pulling it.

At the same time, with the right hand, raise knob No. 4 as high as possible.

The connecting cam is thus released and should come away without effort.

Put the SUPER-8 out of work by pulling out the hook *C*, fig. 42.

Pull out pin 4 and set the knobs for the kind of knitting desired.

Note:

For motifs with an odd number of rows of stitches, the last row from right to left will be knitted with the colour yarn. So that the first row after the pattern is knitted with the ground yarn, knit one tubular row from left to right on the rear, with pin 1 pushed into the cam box.

With the carriage on the right, put the ground yarn carrier in work and pull out pin 1.

Knitting without the Super-8

When the SUPER-8 is not in use, we advise removing the drum, so as to lighten the machine's carriage accordingly. Place the drum carefully in the tool box.

Machines of gauge 25 or 28 (MR or CS)

With the SUPER-8 for machines of gauge 25 and 28 a locking slide is supplied, which fits in the groove of the front guide bar and acts on raising cam No. 1. This locking slide is fitted as follows:

Take the carriage out of the guide bars to the left; insert in the groove of the front guide bar, about 5 cm. from the left end, the small brake shown in fig. 44, with its brass shoe facing upwards.

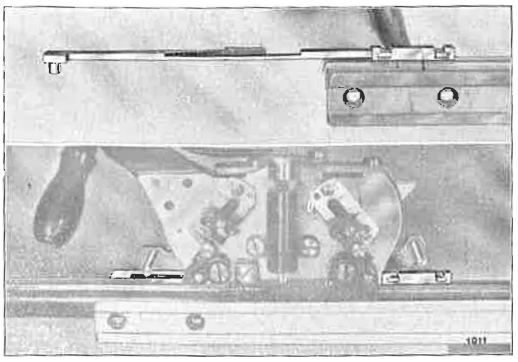


Fig. 44.

Engage the locking slide on the brake. Still holding the slide, put the carriage back in place, resting it in the groove of the front bar opposite the slide, with its right-hand part close to the brake (see fig. 44), then push it into the bars, making the pin at the left end of the slide enter the groove of the front bar. Join the carriage to its connecting rod and put back the left-hand block between the two front bars.

Once this locking slide is in place, it need not be removed for ordinary knitting; however, if the machine is to be used for some time without the SUPER-8, it is better to take it out. To remove it, take the carriage out to the left, so as to free the slide and take out the brake.

Maintenance of the Super-8

Regularly oil the inner faces of the small toothed wheels and of the selector cams. When changing selector cams, take the opportunity to oil the axles.

From time to time put a drop of oil in the lubrication holes of the drum and lightly grease the connecting cam.

Changing selector cams

To remove a selector cam from the drum, wedge it with the small special key supplied with the Super-8. For this, place the key between the cam to be changed and its neighbour on the left,

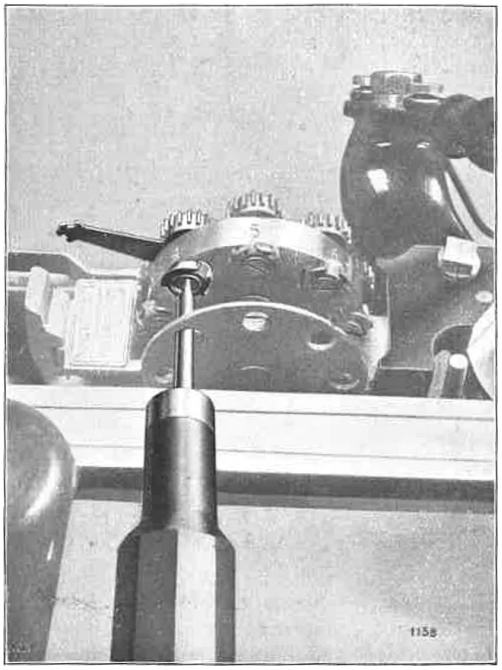


Fig. 45.
Changing a selector cam.

as shown in fig. 45. One side of this key corresponds to gauges 32 and 36, the other side to gauges 25 and 28.

While holding the key with the left hand, unscrew the screw one or two turns with the special screwdriver *, which can pass easily through the holes of the lever. (This screwdriver is supplied with the Super-8.) Try to remove the selector cam; if it does not come, put the screwdriver back through the hole of the lever into the slot of the screw and give a sharp little tap on the end of the screwdriver handle.

Then take the screw and the selector cam right out; hold the small toothed wheel in place, taking care that the stop pin does not escape.

Remove the washer from the selector cam taken out and place it in the new cam, so that the pin forming part of the washer engages in the notch of the cam.

Slide the new selector cam onto the axle of the toothed wheel, matching the pin to the notch of this axle, and push the cam fully onto the axle.

Place the key between the new selector cam and its neighbour on the right.

Put in the screw, tighten it fully and take out the key.

*Important:

Selector cams must always be changed with the special screwdriver, so as not to damage the holes of the SUPER-8 lever.

Cardigan.

Lady's cardigan. Super-8 fronts fig. 49.

Gauge 32. Wool 24/3. 470 grams. Medium size.

The cardigan consists of:

one piece for the back, fig. 46,
two pieces for the sleeves, fig. 47,
two pieces for the fronts, fig. 48,
cardigan borders,
pockets.

Back (fig. 46).

Set the knobs for tubular knitting, that is:

$$\frac{3}{13} \quad \frac{2}{10} \qquad \times \qquad \frac{4}{10} \quad \frac{1}{13}$$

Put 188 needles in work on the front and 188 on the rear.

Cast on and knit 10 tubular cycles.

Set the knobs for 1 × 1 rib, that is:

$$\frac{3}{11\frac{1}{2}} \quad \frac{2}{11\frac{1}{2}}$$
$$\times$$
$$\frac{4}{11\frac{1}{2}} \quad \frac{1}{11\frac{1}{2}}$$

Knit 120 cycles.

On the left, with the transfer tool, transfer 10 stitches of the front bed to the needles of the rear bed. Bring the carriage to the right and with a machine needle make the chain cast-off on these 10 stitches, as shown on page 27.

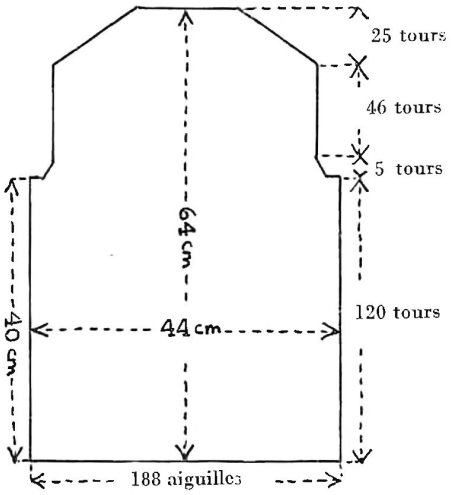


Fig. 46. BACK.

Put the empty needles out of work. Do the same on the right.

The carriage is on the right.

On each side, with the two-point transfer tool, make 1 decrease at 1 cycle, 5 times in succession.

Knit 46 cycles.

On each side, with the two-point transfer tool, make 1 decrease every *row*, 50 times.

Transfer all the stitches of the front bed to the needles of the rear bed and make the chain cast-off as shown on page 15.

Sleeve (fig. 47).

Set the knobs for tubular knitting, that is:

$$\frac{3}{13} \quad \frac{2}{10}$$
$$\times$$
$$\frac{4}{10} \quad \frac{1}{13}$$

Put 94 needles in work on the front and 94 on the rear.

Cast on and knit 45 tubular cycles.

Set the knobs for 1 × 1 rib, that is:

$$\frac{3}{11\frac{1}{2}} \quad \frac{2}{11\frac{1}{2}}$$
$$\times$$
$$\frac{4}{11\frac{1}{2}} \quad \frac{1}{11\frac{1}{2}}$$

Then increase every 4 cycles, 30 times on each side, as follows:

Put one more needle in work, on the front and on the rear in turn, *on the carriage side*, that is:

Carriage on the left. Increase one needle left rear.

Carriage on the right. Increase one needle right front.

Carriage on the left. Increase one needle left front.

Carriage on the right. Increase one needle right rear.

Knit 2 cycles without increasing and repeat *in this way* 30 times *in succession.*

On each side make the chain cast-off on 8 stitches (put the empty needles out of work).

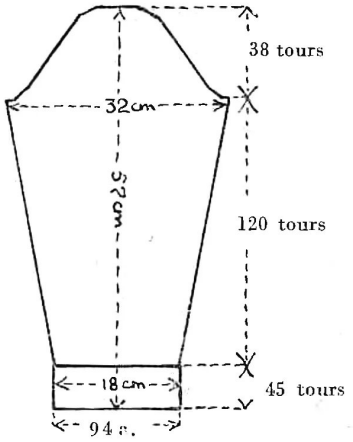


Fig. 47. Sleeve.

Make 3 double decreases at each cycle, then decrease by 1 needle at each cycle, 35 times. Finish with the chain cast-off.

The two sleeves are identical.

Fronts (fig. 48).

Set the knobs for tubular knitting, that is:

3 / 13	2 / 10
4 / 10	1 / 13

Put 102 needles in work on the front and 102 on the rear.

Cast on and knit 10 tubular cycles.

Set the knobs for the Super-8, that is:

3 / 15 free	2 / 12
4 / 4	1 / 12

Place on the indicator the strip of the chosen pattern.

Knit 170 cycles.

On the right, decrease with the two-point transfer tool every 4 cycles, 30 times.

On the left, knit 10 more cycles, make the chain cast-off on 12 stitches, decrease with the two-point transfer tool every cycle 8 times, every 2 cycles 5 times.

Knit 56 cycles, then finish by increasing every 6 cycles, 6 times.

Finish with the chain cast-off.

Make the second piece with the left-hand decreases on the right and the right-hand decreases on the left. This gives one right front and one left front.

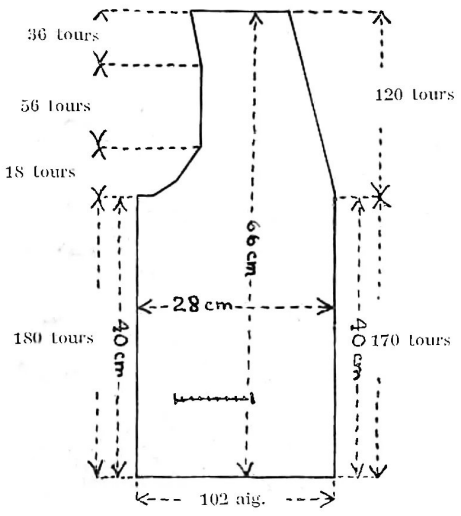


Fig. 48. Front.

Pockets.

Set the knobs for 1 × 1 rib, that is:

3 / 11½	2 / 11½
4 / 11½	1 / 11½

Cast on 46 needles, knit 3 tubular cycles and 32 cycles of 1 × 1 rib, then finish with the chain cast-off.

Make the second pocket identical.

Borders.

Set all 4 knobs to 11.

For the cardigan border, put 16 needles in work and knit 1.90 metres.

For the pocket border, put 9 needles in work and knit 0.30 metres.

Making up.

All the pieces must be pressed along the knitting with a hot iron, placing a damp cloth between the knitted piece and the iron.

Join all the pieces with overcasting stitches (fig. 50). Close the sleeves likewise with overcasting stitches.

Sew the band round the neck opening down to the bottom of the garment, again with small overcasting stitches.

Make 6 buttonholes across the width of the knitting. Cut the buttonhole (fig. 57), take some wool (do not double it), pass it twice through each stitch, then work a close, even buttonhole stitch (fig. 58).

The pockets are set on the inside. First cut the knitting on the right side, taking



Fig. 49. Finished cardigan.

only the top stitches (fig. 57) and following the same row of stitches. Separate the two edges of the slit and remove the strands of wool. Pass a thread through the stitches of the lower edge, which might run.

Sew the edge of the pocket to the upper edge of the slit with overcasting stitches, taking care that the seam shows as little as possible on the outside (preferably by grafting, fig. 52). Sew the other three sides of the pocket with side stitches (fig. 54), parallel to the garment. Finally fit the pocket band with overcasting stitches on the lower edge of the slit and fix it at both ends, not with a turning but simply by passing the needle through each stitch, after unravelling the band until it is exactly the right length.

The buttonholes are made on the right-hand part, the buttons are sewn on the left-hand part.

See the table of cardigan measurements, page 23.

Making up knitted articles

The making up of these articles differs according to whether the pieces have been fashioned to the required shape by decreasing or increasing, or are to be cut.

Fashioned articles.

Joining the various pieces is very easy, for the shapes of knitted garments are generally simple. The selvedges are joined by a seam made by hand in overcasting stitch (fig. 50 and 51). This stitch must be very even, the needle being inserted in each stitch of the selvedge or of the chain cast-off, always at the same depth. For very careful making up, seams running across the width of the knitting are made by grafting (fig. 52), especially on tubular knitting.

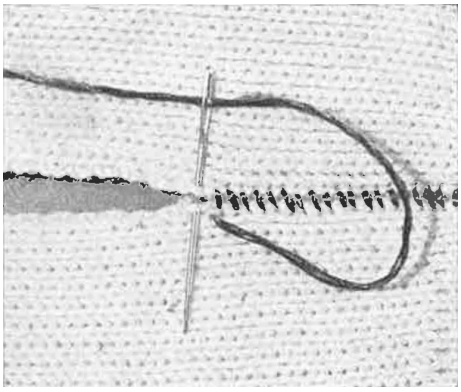


Fig. 50.

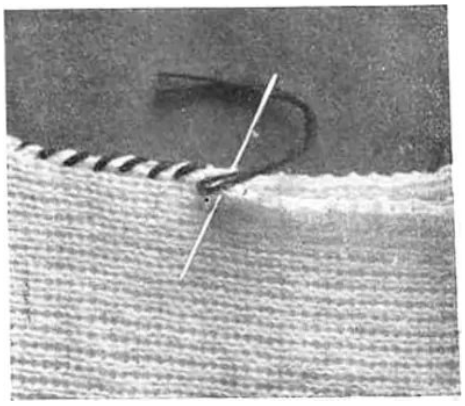


Fig. 51.

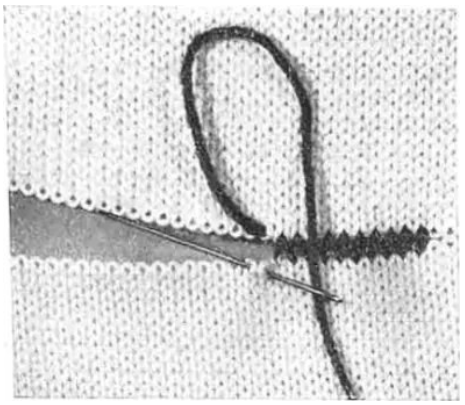


Fig. 52.

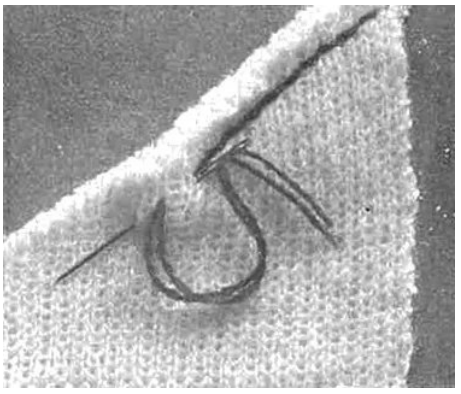


Fig. 53.

Cut articles.

The method of knitting pieces quite straight and then cutting them to the required shape is applied mainly by hosiery manufacturers; it is, however, beginning to spread among home knitters, especially for SUPER-8 articles and for transfer patterns. The loss of material is then small compared with the time saved by knitting without increases or decreases.

When the various pieces are knitted, press them carefully before cutting, so that they are quite flat and the stitches do not run. After cutting the pieces to the required shape, generally with the help of a pattern, join them by hand with very tight backstitches (fig. 53).

Open the seams and cover them with seam tape of a matching shade, unless the knitting is thin enough to allow a felled seam (fig. 54).

Some seams that need not stay elastic can be made on the sewing machine (for example shoulder seams, which must support the knitting and stop it stretching or widening).

Besides the joining proper, there are certain details of making up that must be done carefully, and on which we insist:

Collars and neckbands.

Necklines are almost always cut, whereas collars are finished with

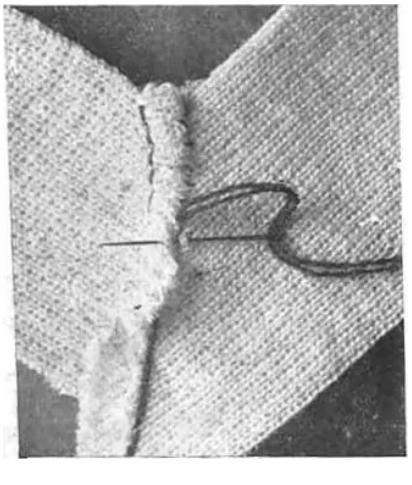


Fig. 54.

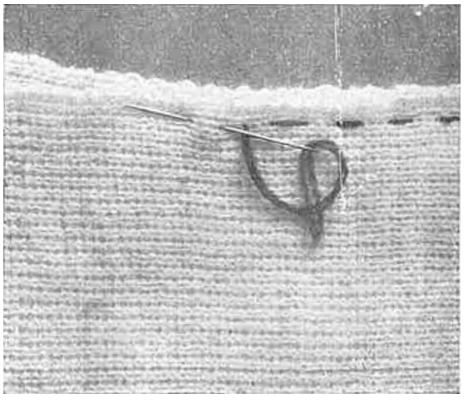


Fig. 55.

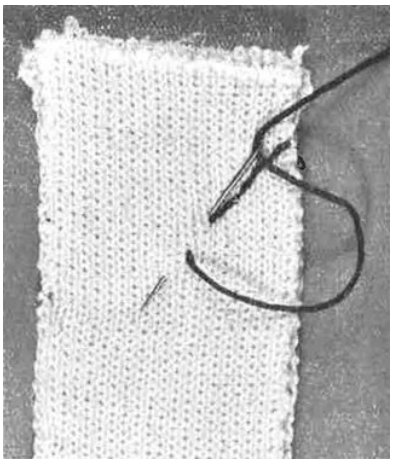


Fig. 56.

a chain cast-off. Fit the collar on the inside with a running stitch (fig. 55), and turn the cut knitting over to the outside (under the collar) with a hem stitch (fig. 54).

The bands of pullovers are fitted in different ways: either single, laid flat on the knitting with running stitches (the knitting is then turned over to the wrong side on the band), or double (astride the knitting). This last method is used mainly when the band is in tubular knitting made across the full width of the machine. It must then be fitted with running stitches, on the wrong side and on the right side, inserting the needle in each stitch.

In all cases, sew the point before fitting the band (fig. 56), otherwise it cannot be made exactly.

Pockets.

The pockets are set on the inside. First cut the knitting, on the right side, taking only the top stitches (see fig. 57) and following the same row of stitches. If the knitting is in 1 × 1 rib, one need only separate the two edges of the slit and remove the strands of wool. In other kinds of knitting, cut the stitches on the wrong side. Pass a thread through the stitches of the lower edge, which might run.

Sew the edge of the pocket to the upper edge of the slit with overcasting stitches, taking care that the seam shows as little as possible on the outside. Sew the other three sides of the pocket with running stitches, parallel to the garment. Finally fit the pocket band with overcasting stitches on the lower edge of the slit and fix it at both ends, not with a turning but simply by passing the needle through each stitch, after unravelling the band until it is exactly the right length.

Buttonholes.

Buttonholes are made as far as possible across the width of the knitting. Cut the buttonhole in the same way as the pocket slit. Take fine wool (do not double it), cotton or silk according to the knitted garment, pass a thread through the stitches, twice, then work a close, even buttonhole stitch (see fig. 58).

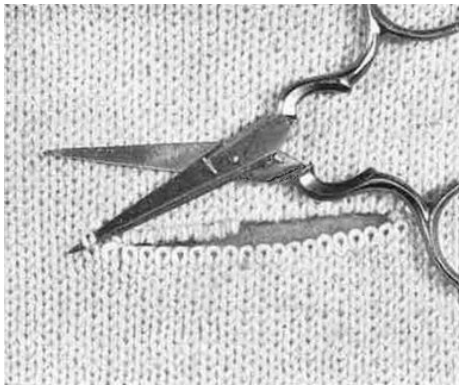


Fig. 57.

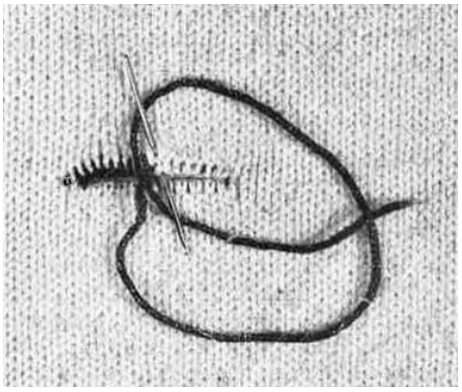


Fig. 58.

Stockings and socks.

STOCKINGS in plain tubular knitting, normal ladies' size.

Gauge 32. Wool 3/24 decatized. Weight of the pair: 130 grams.

Length of the leg (welt sewn) 62 cm.
Length of the foot 26 cm.

Knob settings:

Knobs 1 and 3 at 14 for the cast-on and the welt.

Knobs 1 and 3 at 13 for the rest of the stocking.

Knobs 2 and 4 at 10.

Leg.

Put 70 needles in work on the front and 70 on the rear.

Cast on, close pins 1 and 3 and knit 50 cycles for the welt.

Set knobs 1 and 3 at 13.

Knit 180 cycles. (To avoid mistakes, use the counter.)

Stop the carriage on the left.

With the three-point transfer tool, decrease 1 needle on the right on the front and on the rear. Put the 2 empty needles out of work.

Knit 10 cycles.

Make one decrease on the right, as above.

Knit 9 cycles. Make one decrease on the right.
" 8 " " " " "
" 7 " " " " "
" 6 " " " " "

Make 21 more decreases on the right, with 5 cycles between them.

There will thus be 26 decreases in all for the calf.

After the last decrease, knit 60 cycles.

Stop the carriage on the right. Close pins 2 and 4.

Bring the carriage to the left, without a yarn carrier.

The leg is now finished. There remain 44 needles on the front and on the rear.

To knit the foot, the leg is taken off and put back on the machine so that the decreases are in the middle on the rear. For this, take off the weights and the cast-on comb, take 4 stitch holders with 24 points, place two on the front and two on the rear, passing all the stitches onto these holders. The 2 front holders are placed so that they have 2 empty needles on the left, the rear ones so that they have 2 empty needles on the right. Proceed for this operation as when using the transfer tool (see fig. 33 to 35, page 30). Fit the covers on the holders.

Lower the front bed.

Pass the 4 holders under the machine, between the beds, without breaking the yarn.

Put back in work on the right, on the front and on the rear, the needles up to 22, and put out of work on the left, on the front and on the rear, the needles up to 22. There will thus be 44 needles in work on the front and on the rear. Open the latches of all the rear needles.

Turn the leg and the holders so that the decreases are in the middle on the rear bed.

Bring the two rear holders up between the two beds. Take off their covers and pass the stitches onto the needles.

The two front holders stay hanging between the two beds.

Raise the front bed.

Bring the yarn carrier to the left and undo the stitches by hand from the middle to the left, so that the yarn is at the left-hand selvedge.

Heel.

Open pins 2 and 3.

Set the knobs:

$$\frac{3}{13}\ \frac{4}{0} \quad \times \quad \frac{2}{13}\ \frac{1}{0}$$

Hook on the heel claw with a small weight in the middle at the rear.

Add if desired a reinforcing yarn or "single end", passing it through the same yarn carrier as the main yarn.

Knit 18 cycles for the heel flap.

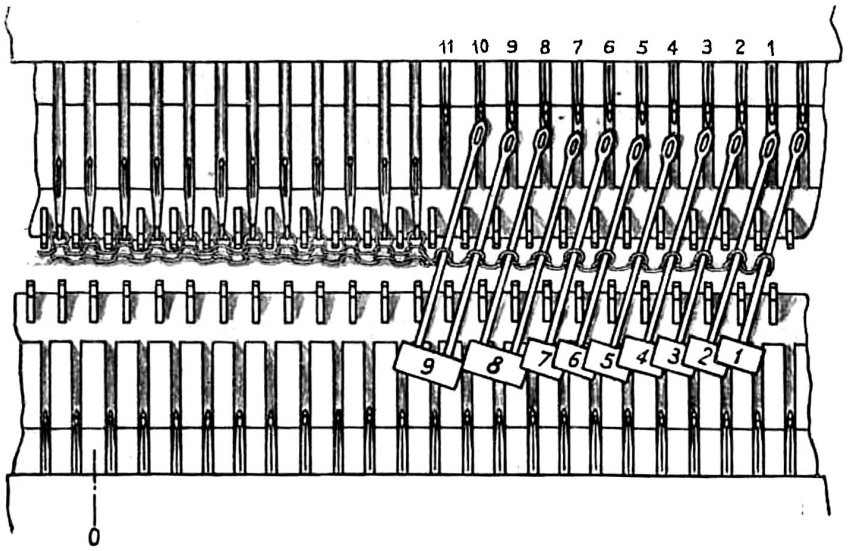


Fig. 59. Stitch holding.

Stop the carriage on the left.

On the right place 7 one-point stitch holders and 2 two-point stitch holders, in other words place the first 7 stitches each on a one-point holder and the next 4 stitches on 2 two-point holders.

Place the holders astride the two beds (see fig. 59).

Put all the empty needles out of work.

Bring the carriage to the right.

On the left place 7 one-point holders and 2 two-point holders, as was just done on the right.

With the two-point holders near the selvedge, transfer one stitch to the selvedge needle and one stitch to the first empty needle. Put the other 10 empty needles on the left out of work.

Bring the carriage to the left.

Put one needle in work on the right, and transfer to this needle and to the selvedge needle the two stitches of the first two-point holder.

As the stitches of the holders are transferred, move all the holders closer to the selvedge.

Bring the carriage to the right.

Put one needle in work on the left and transfer the stitches of the second two-point holder as was done with those of the first holder.

Bring the carriage to the left and do the same on the right as above.

Bring the carriage to the right and place the stitch of the first holder on the left on the selvedge needle.

Bring the carriage to the left, do the same on the right as above, and so on, left and right alternately, until there are no holders left.

The last row will bring the carriage back to the left.

Open the latches of all the front needles.

Lower the front bed and bring the claw and the weight up close to the rear needles.

Bring up the two front holders, take off their covers and pass the stitches onto the front needles.

Put the front bed back in place.

Put 4 needles in work on the right on the front, and on the rear the 14 corresponding needles.

Lower the front bed.

Hook on to the empty needles the small selvedge stitches (knots) of the heel flap. Take care at the same time that the heel stitches do not come off the needles.

Begin by transferring the first knot to

the front selvedge needle and continue on the 4 front needles, then on the rear.

Unscrew knob 2. Put the front bed back in place. Pull the knitting by hand and bring the carriage to the right.

Put in work on the left 5 needles on the front bed and the 13 corresponding needles on the rear bed.

On the rear, place the first chain stitch on the selvedge needle, then pick up on the left the selvedge stitches of the heel flap, as was just done on the right.

Hook on a claw with a small weight on each side of the heel. Hang the edge-weight loop from the leg of the stocking with a medium weight.

Open pin No. 4 and close pin No. 3.

Remove the reinforcing yarn.

Set knob No. 1 at 13, knob No. 2 at 10 and unscrew knob No. 3.

Bring the carriage to the left.

Set knob No. 3 at 13 and knob No. 4 at 10.

Heel gusset.

Knit 3 cycles.

Make one decrease on the front, right and left, with the three-point transfer tool. Lower the empty needles.

Knit 3 cycles.

Make one decrease on the front, right and left, with the two-point tool, then with the one-point tool transfer to the empty front needle the rear selvedge stitch. Lower the empty needles on the rear, left and right.

Knit 3 cycles.

Make one decrease on the front, right and left, with the two-point tool. Lower the empty needles.

Knit 3 cycles.

Make one decrease on the front, right and left, with the one-point tool and transfer to the empty needle the rear selvedge stitch. Lower the empty needles on the rear.

Knit 3 cycles.

Make one decrease on the front, right and left, with the one-point tool. Lower the empty needles.

Knit 3 cycles.

Make one decrease on the front, right and left, with the one-point tool and transfer to the empty needle the rear selvedge stitch. Lower the empty needles on the rear.

Note: Some knitters form the heel gusset simply by decreasing on the front and on the rear, but the shape of the foot is not as exact and the appearance not as fine as when the gusset is made by the method given above.

Knit 85 cycles for the length of the foot.

Toe.

Add the reinforcing yarn again if desired.

Decrease on each side, on the front and on the rear, with the three-point transfer tool, 8 times with 3 cycles between.

Lower the empty needles regularly as the decreases proceed.

Remove the medium weight and leave only the two claws with the small weights.

Then decrease as above, 5 times with 2 cycles between, then 8 times with one cycle between, then every cycle.

As soon as the number of needles is down to 5, use the one-point tool to finish the decreases.

When only 2 needles remain on the front and one on the rear, take off the weights, transfer the two stitches of the front needles to the rear needle and lower the front needles.

Pull the yarn below the yarn-carrier tip, between the two beds, break it and secure it under the machine.

Pass the end of the yarn through the last stitch to finish the work.

Finishing the stocking.

Press the top of the stocking with a damp cloth so that it does not roll. Cut the first and last stitches of the net and pull the yarn to open the stocking. Fold the knitting inside and form the welt by sewing it, stitch by stitch.

Other sizes.

For the other stocking sizes, in gauges 32 and 36, refer respectively to tables *A* and *B*, pages 50 and 51.

Socks.

To make socks, first knit the ribbed border, either in *1* × 1 rib or in 2 × 2 rib, then transfer it to the machine to knit the foot.

In general the ribbed border has a greater number of stitches than the tubular part; this must be allowed for when transferring it to the machine to knit the leg and foot. For example, a 1 × 1 ribbed border for a medium-size gauge 32 sock has 120 stitches (60 on the front and 60 on the rear): it is transferred to 84 needles to knit the leg.

There is therefore an excess of 36 stitches to be distributed evenly over the 2 beds; the diagram below shows the distribution of 18 stitches on one of the beds. The needles marked with a * carry two stitches.

I I I

Table "A" Stockings in plain knitting (tubular)

Gauge 32 (8) — Wool 3/24 metric — Knobs 1 & 3 { at 14 for the welt. at 13 for the rest of the stocking.

SIZES	Number of needles front and rear for the cast-on	Number of cycles for the welt	Number of cycles up to the 1st decrease	Number of decreases	Number of cycles between the decreases	Number of cycles before the heel	Number of needles front and rear for the foot	Number of cycles for the heel flap	Number of stitch holders on each side	Number of needles to put in work on the front to pick up the chains	Number of decreases for the heel gusset and cycles between	Number of cycles for the foot	Number of decr. for the toe and cycles between	Length of the leg in cm	Length of the foot in cm	Approx. weight per pair in gr.
up to 3 months	34	30	40	6	5 cycles 4 cycles 4×3 cycles	30	30	15	3 at 1 nd. 2 at 2 nd.	3	3 at 5 cy.	32	6 at 2 cy. 8 at 1 cy.	18	11	30
up to 1 year .	38	30	55	6	6 cycles 5 cycles 4×4 cycles	30	32	15	3 at 1 nd. 2 at 2 nd.	3	3 at 5 cy.	40	6 at 2 cy. 8 at 1 cy.	22	12	40
2 years	42	40	70	10	7 cy, 6 cy, 5 cy, 5×4 cycles	40	32	15	4 at 1 nd. 2 at 2 nd.	3	3 at 6 cy.	45	6 at 2 cy. 8 at 1 cy.	26	14	50
3 and 4 years ..	46	45	85	12	7 cy, 6 cy, 5 cy, 9×4 cycles	40	34	16	4 at 1 nd. 2 at 2 nd.	3	3 at 6 cy.	50	7 at 2 cy. 8 at 1 cy.	30	16	60
5 and 6 years ..	50	45	100	16	8 cy, 7 cy, 6 cy, 5 cy, 12×4 cycles	40	34	16	4 at 1 nd. 2 at 2 nd.	3	3 at 6 cy.	55	7 at 2 cy. 8 at 1 cy.	35	17	70
7 and 8 years ..	54	45	110	18	8 cy, 7 cy, 6 cy, 5 cy, 14×4 cycles	45	36	16	5 at 1 nd. 2 at 2 nd.	3	3 at 6 cy.	58	2 at 3 cy. 5 at 2 cy. 8 at 1 cy.	40	18	80
9 and 10 years .	58	50	125	20	9 cy, 8 cy, 7 cy, 6 cy, 16×5 cycles	45	38	17	5 at 1 nd. 2 at 2 nd.	4	3 at 6 cy.	62	2 at 3 cy. 5 at 2 cy. 8 at 1 cy.	45	19	90
11 and 12 years	60	50	140	22	9 cy, 8 cy, 7 cy, 6 cy, 18×5 cycles	50	38	17	6 at 1 nd. 2 at 2 nd.	4	3 at 6 cy.	68	3 at 3 cy. 5 at 2 cy. 8 at 1 cy.	50	21	100
13 and 14 years	64	50	150	24	9 cy, 8 cy, 7 cy, 6 cy, 19×5 cycles	50	40	18	6 at 1 nd. 2 at 2 nd.	4	3 at 6 cy.	75	3 at 3 cy. 5 at 2 cy. 8 at 1 cy,	55	22	110
15 and 16 years	68	50	160	24	10 cy, 9 cy, 8 cy, 7 cy, 6 cy, 20×5 cycles	60	44	18	6 at 1 nd. 2 at 2 nd.	4	3 at 6 cy.	80	5 at 3 cy. 5 at 2 cy. 8 at 1 cy.	60	24	120
17 years and over	70	50	180	26	10 cy, 9 cy, 8 cy, 7 cy, 6 cy, 21×5 cycles	60	44	18	7 at 1 nd. 2 at 2 nd.	4	3 at 6 cy.	85	8 at 3 cy. 5 at 2 cy. 8 at 1 cy.	65	26	130

Table "B".

Stockings in plain knitting (tubular)

Gauge 36 (7) — Wool 3/20 metric — Knobs 1 & 3 { at 16 for the welt. / at 15 for the rest of the stocking.

SIZES	Number of needles front and rear for the cast-on	Number of cycles for the welt	Number of cycles up to the 1st decrease	Number of decreases	Number of cycles between the decreases	Number of cycles before the heel	Number of needles front and rear for the foot	Number of cycles for the heel flap	Number of stitch holders on each side	Number of ndl. to put in work on the front to pick up the chains	Number of decreases for the heel gusset and cycles between	Number of cycles for the foot	Number of decr. for the toe and cycles between	Length of the leg in cm	Length of the foot in cm	Approx. weight per pair in gr.
up to 3 months	30	30	35	6	5 cycles 4 " 4×3 cycles	25	24	14	3 at 1 nd. 2 at 2 nd.	3	3 at 4 cy.	25	2 at 2 cy. 8 at 1 cy.	18	11	40
up to 1 year .	34	30	50	6	5 cycles 4 " 4×3 cycles	25	28	14	3 at 1 nd. 2 at 2 nd.	3	3 at 4 cy.	28	3 at 2 cy. 8 at 1 cy.	22	12	50
2 years	38	35	65	8	6 cy, 5 cy, 4 cy, 5×3 cycles	30	30	14	4 at 1 nd. 2 at 2 nd.	3	3 at 5 cy.	32	3 at 2 cy. 8 at 1 cy.	26	14	60
3 and 4 years ..	42	35	80	10	7 cy, 6 cy, 5 cy, 7×4 cycles	30	32	15	4 at 1 nd. 2 at 2 nd.	3	3 at 5 cy.	36	4 at 2 cy. 8 at 1 cy.	30	16	70
5 and 6 years ..	44	35	90	10	7 cy, 6 cy, 5 cy, 7×4 cycles	35	34	15	4 at 1 nd. 2 at 2 nd.	3	3 at 5 cy.	40	2 at 3 cy. 5 at 2 cy. 8 at 1 cy.	35	17	80
7 and 8 years ..	46	35	100	12	8 cy, 7 cy, 6 cy, 9×5 cycles	40	34	15	4 at 1 nd. 2 at 2 nd.	3	3 at 5 cy.	44	2 at 3 cy. 5 at 2 cy. 8 at 1 cy.	40	18	90
9 and 10 years .	48	40	110	13	8 cy, 7 cy, 6 cy, 10×5 cycles	40	35	16	5 at 1 nd. 2 at 2 nd.	4	3 at 5 cy.	48	3 at 3 cy. 5 at 2 cy. 8 at 1 cy.	45	19	100
11 and 12 years	50	40	120	14	9 cy, 8 cy, 7 cy, 6 cy, 10×5 cycles	45	36	16	5 at 1 nd. 2 at 2 nd.	4	3 at 5 cy.	52	3 at 3 cy. 5 at 2 cy. 8 at 1 cy.	50	21	110
13 and 14 years	52	40	130	16	9 cy, 8 cy, 7 cy, 6 cy, 12×5 cycles	45	36	17	5 at 1 nd. 2 at 2 nd.	4	3 at 5 cy.	56	3 at 3 cy. 5 at 2 cy. 8 at 1 cy.	55	22	120
15 and 16 years	56	40	145	18	10 cy, 9 cy, 8 cy, 7 cy, 6 cy, 13×5 cycles	50	38	17	6 at 1 nd. 2 at 2 nd.	4	3 at 5 cy.	60	4 at 3 cy. 5 at 2 cy. 8 at 1 cy.	60	24	130
17 years and over	60	40	170	19	10 cy, 9 cy, 8 cy, 7 cy, 6 cy, 14×5 cycles	50	42	17	6 at 1 nd. 2 at 2 nd.	4	3 at 5 cy.	64	4 at 3 cy. 5 at 2 cy. 8 at 1 cy.	65	26	140

Table "C"

Men's wool socks, with 1 × 1 ribbed border

GAUGE 32 (8) — Wool 3/24 metric — Knobs { Nos. 1 and 3 at 14 for the cast-on. Nos. 1 to 4 at 12 for the ribbed border. Nos. 1 and 3 at 13 for the leg and foot.

SIZES	Number of needles front and rear for the ribbed border	Number of tubular cycles for the welt	Number of cycles for the ribbed border	Length of the ribbed border in cm	Number of needles front and rear for the leg	Number of cycles for the leg	Number of cycles for the heel flap	Number of stitch holders on each side	Number of needles to put in work on the front to pick up the chains	Number of decreases for the heel gusset and cycles between	Number of cycles for the foot	Number of decreases for the toe and cycles between	Length of the foot in cm	Weight per pair, in wool	Weight per pair, in cotton
Small	58	3	40	13	46	95	20	7 at 1 needle 2 at 2 needles	4	4 at 5 cycles	90	5 at 3 cycles 8 at 2 cycles 9 at 1 cycle	27	90	80
Medium	60	3	40	13	48	100	21	7 at 1 needle 2 at 2 needles	5	4 at 6 cycles	95	5 at 3 cycles 8 at 2 cycles 10 at 1 cycle	28	90	90
Large	62	3	40	13	50	105	22	8 at 1 needle 2 at 2 needles	6	5 at 6 cycles	100	6 at 3 cycles 8 at 2 cycles 11 at 1 cycle	29	90	90

Table "D"

Men's wool socks, with leg in 2 × 2 rib

GAUGE 32 (8) — Wool 3/24 metric — Knobs { Nos. 1 and 3 at 14 for the cast-on. Nos. 1 to 4 at 13 for the 2 × 2 rib.

SIZES	Number of ribs front and rear	Number of tubular cycles for the welt	Number of cycles for the leg	Length of the leg in cm	Number of needles front and rear for the foot	Number of tubular cycles	
Small	22	3	75	25	46	5	then as above in table "C".
Medium	24	3	75	25	48	5	
Large	26	3	75	25	50	5	

Table “E”. Men’s wool socks, with 1 × 1 ribbed border

GAUGE 36 (7) — Wool 3/20 metric — Knobs { Nos. 1 and 3 at 15 for the cast-on. Nos. 1 to 4 at 13 for the ribbed border. Nos. 1 and 3 at 14 for the leg and foot.

SIZES	Number of needles front and rear for the ribbed border	Number of tubular cycles for the welt	Number of cycles for the ribbed border	Length of the ribbed border in cm	Number of needles front and rear for the leg	Number of cycles for the leg	Number of cycles for the heel flap	Number of stitch holders on each side	Number of needles to put in work on the front to pick up the chains	Number of decreases for the heel gusset and cycles between	Number of cycles for the foot	Number of decreases for the toe and cycles between	Length of the foot in cm	Weight per pair
Small	52	3	35	13	42	85	18	7 at 1 needle 2 at 2 needles	4	3 at 5 cycles	75	5 at 3 cycles 8 at 2 cycles 8 at 1 cycle	27	90
Medium	54	3	35	13	44	90	19	7 at 1 needle 2 at 2 needles	4	3 at 6 cycles	80	5 at 3 cycles 8 at 2 cycles 9 at 1 cycle	28	90
Large	56	3	35	13	46	95	20	8 at 1 needle 2 at 2 needles	5	4 at 6 cycles	85	6 at 3 cycles 8 at 2 cycles 9 at 1 cycle	29	90

Table “F”. Men’s wool socks, with leg in 2 × 2 rib

GAUGE 36 (7) — Wool 3/20 metric — Knobs { Nos. 1 and 3 at 15 for the cast-on. Nos. 1 to 4 at 14 for the 2 × 2 rib.

SIZES	Number of ribs front and rear	Number of tubular cycles for the welt	Number of cycles for the leg	Length of the leg in cm	Number of needles front and rear for the foot	Number of tubular cycles	
Small	20	3	65	25	42	5	then as above in table “E”.
Medium	22	3	65	25	44	5	
Large	24	3	65	25	46	5	

Table "G". Men's cotton socks, with 1 × 1 ribbed border

GAUGE 32 — Cotton 30/6 — Knobs { Nos. 1 and 3 at 15 for the cast-on; Nos. 1 to 4 at 11½ for the ribbed border; Nos. 1 and 3 at 14 for the leg and foot

SIZES	Number of needles front and rear for the ribbed border	Number of tubular cycles for the welt	Number of cycles for the ribbed border	Length of the ribbed border in cm	Number of needles front and rear for the leg	Number of cycles for the leg	Number of cycles for the heel flap	Number of stitch holders on each side	No. of ndl. to put in work on the front to pick up the chains	Number of decreases for the heel gusset and cycles between	Number of cycles for the foot	Number of decreases for the toe and cycles between	Length of the foot in cm.	Weight per pair
Small . . .	64	3	45	12	48	105	20	7 at 1 needle 2 at 2 needles	4	4 at 4 cycles	90	3 at 3 cycles 6 at 2 cycles the rest at 1 cycle	27	80
Medium . .	68	3	45	12	49	105	21	8 at 1 needle 2 at 2 needles	5	5 at 5 cycles	94	4 at 3 cycles 7 at 2 cycles the rest at 1 cycle	28	80
Large . . .	70	3	45	12	50	105	21	8 at 1 needle 2 at 2 needles	5	5 at 5 cycles	98	5 at 3 cycles 8 at 2 cycles the rest at 1 cycle	29	80

Table "H". Men's cotton socks, with leg in 2 × 2 rib

GAUGE 32 — Cotton 30/6 — Knobs { Nos. 1 and 3 at 15 for the cast-on; Nos. 1 to 4 at 12 for the leg

SIZES	Number of ribs front and rear	Number of tubular cycles for the welt	Number of cycles for the leg	Length of the leg in cm.	Number of needles front and rear for the foot	Number of tubular cycles	
Small . . .	25	3	85	25	48	3	then as above in table No. G.
Medium . .	27	3	85	25	49	3	
Large . . .	29	3	85	25	50	3	

Table "J". Men's cotton socks, with 1 × 1 ribbed border

GAUGE 36 – Cotton 30/8 – Knobs { Nos. 1 and 3 at 16 for the cast-on
Nos. 1 to 4 at 12 for the 1 × 1 rib
Nos. 1 and 3 at 14½ for the leg and foot

SIZES	Number of needles front and rear for the ribbed border	Number of tubular cycles for the welt	Number of cycles for the ribbed border	Length of the ribbed border in cm	Number of needles front and rear for the leg	Number of cycles for the leg	Number of cycles for the heel flap	Number of stitch holders on each side	No. of ndl. to put in work on the front to pick up the chains	Number of decreases for the heel gusset and cycles between	Number of cycles for the foot	Number of decreases for the toe and cycles between	Length of the foot in cm.	Weight per pair
Small . . .	56	3	35	12	42	95	20	7 at 1 needle 2 at 2 needles	4	4 at 4 cycles	86	3 at 3 cycles 5 at 2 cycles the rest at 1 cycle	27	80
Medium . .	58	3	35	12	44	95	20	7 at 1 needle 2 at 2 needles	4	4 at 5 cycles	90	4 at 3 cycles 6 at 2 cycles the rest at 1 cycle	28	80
Large . . .	60	3	35	12	46	95	21	8 at 1 needle 2 at 2 needles	5	5 at 5 cycles	94	5 at 3 cycles 7 at 2 cycles the rest at 1 cycle	29	80

Table "K". Men's cotton socks, with leg in 2 × 2 rib

GAUGE 36 – Cotton 30/8 – Knobs { Nos. 1 and 3 at 16 for the cast-on
Nos. 1 to 4 at 12 for the leg

SIZES	Number of ribs front and rear	Number of tubular cycles for the welt	Number of cycles for the leg	Length of the leg in cm.	Number of needles front and rear for the foot	Number of tubular cycles	
Small . . .	23	3	80	25	42	3	then as above in table No. J.
Medium . .	25	3	80	25	44	3	
Large . . .	27	3	80	25	46	3	

Sports stockings.

Super-8 sports stockings with border in 2 × 2 rib or tubular knitting (Large size).

Gauge 32. Wool 24/3. 160 grams the pair.



Fig. 60. Sports stocking.

LEG.

For the border in 2 × 2 rib, arrange 108 needles on the front and 108 needles on the rear as follows:

I . I I . I I . I I . I I . I I . I rear
I I . I I . I I . I I . I I . I I front

Knobs at $\frac{3}{13\frac{1}{2}}$ $\frac{2}{10}$ × $\frac{4}{10}$ $\frac{1}{13\frac{1}{2}}$

Rack, cast on and knit 3 tubular cycles, then rack back and knit 40 cycles of 2 × 2 rib.

For the border in tubular knitting, put 108 needles in work on the front and 108 on the rear.

Knobs at $\frac{3}{13\frac{1}{2}}$ $\frac{2}{10}$ × $\frac{4}{10}$ $\frac{1}{13\frac{1}{2}}$

Cast on and knit 80 tubular cycles.

Put the Super-8 in work.

Knobs at $\frac{3}{15\text{ free}}$ $\frac{2}{13}$ × $\frac{4}{4}$ $\frac{1}{12}$

Choose any pattern. Knit 32 cycles.

Decrease on each side with the two-point transfer tool, every 6 cycles, 15 times.

Knit 40 cycles.

On the left and on the right, transfer 18 stitches of the rear bed to 18 needles of the front bed. Then put the 18 double stitches of the front bed on stitch holders. Put the empty needles out of work.

Go on knitting, still with the Super-8, on the 42 remaining stitches, for 78 cycles.

Transfer the stitches of the rear bed to the needles of the front bed, then put these double stitches on stitch holders.

HEEL FLAP AND HEEL.

Put 44 needles in work on the front and 44 on the rear. To begin the heel, hook on the rear the 36 double stitches previously taken on stitch holders, then split a few stitches to reach the figure 44.

Knobs 2 and 3 at 14, and knobs 1 and 4 at 0.

Pins 2 and 3 open and pins 1 and 4 closed.

Take the wool double and knit 17 cycles.

The carriage is on the left.

On the right, put 11 stitches on 11 one-point stitch holders.

Put the empty needles out of work. Bring the carriage to the right. On the left, put 11 stitches on 11 one-point stitch holders. Put the empty needles out of work. Bring the carriage to the left.

On the right, transfer the first stitch put on a holder to the last needle left in work. Bring the carriage to the right.

On the left, transfer the first stitch put on a holder to the last needle left in work. Bring the carriage to the left, and so on, until there are no stitches left on the holders.

The carriage is on the left.

On the right, put 17 needles in work to hook on them the 17 chains of the heel flap. Bring the carriage to the right.

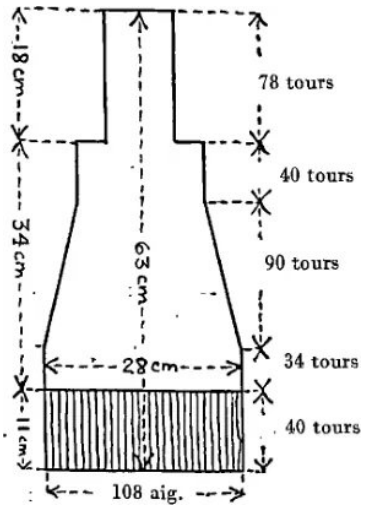


Fig. 61. Leg.

On the left, put 17 needles in work to hook on them the 17 chains of the heel flap. Bring the carriage to the left.

Knit 3 cycles. With the two-point transfer tool, decrease one needle on the left and on the right. Repeat this 5 times, always knitting 3 cycles between.

Knit 40 cycles.

TOE.

For the toe, hook on the front bed the 40 double stitches previously taken on stitch holders, then split a few stitches to reach the figure 44.

Knobs at $\frac{3}{14}$ $\frac{2}{10}$ × $\frac{4}{10}$ $\frac{1}{14}$

Pins 1 and 3 closed, pins 2 and 4 open.

Knit 3 cycles. On the left and on the right, on the front and on the rear, with the two-point transfer tool, make 4 decreases at 3 cycles, then 4 decreases at 2 cycles, then at 1 cycle, until only 4 stitches remain. Knit a few cycles, take the work off the machine and finish the toe by hand.

MAKING UP.

Sew the whole leg on the wrong side with overcasting stitches, inserting the needle in each stitch. Then make the two seams of the foot, also with overcasting. Carefully secure all the yarn ends.

Table “L”. Children’s sports stockings. Wool 20/3.

Gauge 32 - No. 8

Knobs: 1 × 1 rib $\frac{12\ 12}{12\ 12}$ heel and sole of the foot $\frac{15\ 15}{10\ 10}$ toe $\frac{15\ 10}{10\ 15}$

SIZES	Number of needles front and rear for the cast-on	Number of cycles for the border	Number of cycles up to the 1st decrease	Number of decreases	Number of cycles between the decreases	Number of cycles before the heel	Number of stitches to put on stitch holders	Number of cycles for the top of the foot	Number of needles for the heel	Number of cycles for the heel flap	Number of stitch holders on each side	Number of needles to put in work	Number of decr. for the heel gusset and cycles between	Number of cycles for the foot	Number of needles for the toe	Number of decr. and cycles for the toe	Length of the foot in cm	Length of the leg in cm
4	62	13	26	7	4×3 cycles 3×2 cycles	15	2×12	32	24	13	3 at 1 2 at 2	13	8×3 cycles	6	24	4×2 cycles the rest at 1 cycle	15	16
5	69	14	30	7	4×3 cycles 3×2 cycles	15	2×14	35	28	14	3 at 1 2 at 2	14	9×3 cycles	6	27	4×2 cycles, the rest at 1 cycle	16	19
6	77	15	34	7	7×3 cycles	15	2×15	38	32	15	4 at 1 2 at 2	15	7×4 cycles	8	31	4×2 cycles, the rest at 1 cycle	18	22
7	83	15	38	8	8×3 cycles	15	2×17	40	34	15	5 at 1 2 at 2	15	8×4 cycles	6	33	5×2 cycles, the rest at 1 cycle	20	24

Gauge 36 - No. 7

Knobs: 1 × 1 rib $\frac{13\ 13}{13\ 13}$ heel and sole of the foot $\frac{16\ 16}{10\ 10}$ toe $\frac{16\ 10}{10\ 16}$

4	56	12	24	6	6×2 cycles	13	2×11	30	22	12	3 at 1 2 at 2	12	7×3 cycles	5	22	4×2 cycles, the rest at 1 cycle	15	15
5	63	13	28	6	6×2 cycles	14	2×13	33	26	13	3 at 1 2 at 2	13	8×3 cycles	5	26	4×2 cycles, the rest at 1 cycle	16	18
6	72	14	35	6	4×3 cycles 2×2 cycles	15	2×14	35	28	14	4 at 1 2 at 2	14	6×4 cycles	5	28	4×2 cycles, the rest at 1 cycle	17	21
7	78	14	38	7	5×3 cycles 2×2 cycles	16	2×15	37	30	14	4 at 1 2 at 2	14	7×5 cycles	3	30	5×2 cycles, the rest at 1 cycle	19	23

Care and running of the machine.

To look after the machine, remove the dust from the guide bars and the carriage every evening with a rag and a brush, then every morning put a little oil on the guide bars of the yarn carriers and the carriage.

Do not forget to grease from time to time the sloping faces of the ring that drives the counter, the supports of the crank axle and the parts of the stop blocks in contact with the driving plunger of the yarn carriers.

Regularly and carefully remove the build-ups of material that form after some time at work in the entries of the bed blocks, on the lowering brackets, under the tilting cams, etc.

Take special care that no drop of oil ever falls on the needle field.

Always use the special DUBIED oil.

Table of yarn counts to use on the usual gauges of Dubied flat knitting machines.

Gauges	No. (needles per English inch)	WOOL		COTTON English	SILK International (denier)
		Metric	English Bradford		
25	10	2/25. 2/28, 2/30 2/32, 2/36	2/22 at 2/32	4/20, 3/20, 4/30 2/20	450 at 800
28	9	2/20, 2/25, 2/28 2/30, 2/32	2/18 at 2/26	5/20, 4/20, 3/20 4/30	500 at 1000
32	8	2/16, 3/25, 2/20 2/25	2/14 at 2/22	6/20, 5/20, 4/20 6/30, 4/30	600 at 1200
36	7	2/12, 2/16, 3/25 2/20. 3/30	2/11 at 2/16	8/20, 7/20, 6/20 8/30, 5/20. 6/30	750 at 1500

Type M machine.

This small machine, suited mainly to making narrow articles such as stockings, socks or children's garments, is generally found 36 cm wide, gauge 36, or 31 cm wide, gauges 21, 25, 28 or 32.

It has only one yarn carrier, mounted directly on the rear plate of the carriage (see fig. 62).

So that it takes up a correct position at each pass of the carriage, this yarn carrier is joined to a brake A sliding on a steel bar B.

Do not forget to oil this bar often.

The M machine 31 cm wide and above has on the front, besides the usual stitch cams, adjustable auxiliary stitch cams, whose use is given on page 61.

The M machine is generally supplied with a lighter table than the MM table. The machine is fixed directly to the wooden top by means of the clamping brackets (read the paragraphs in italics on page 9).

The M machine can take the plating tip, but neither the push-button yarn changer nor the Super-8.

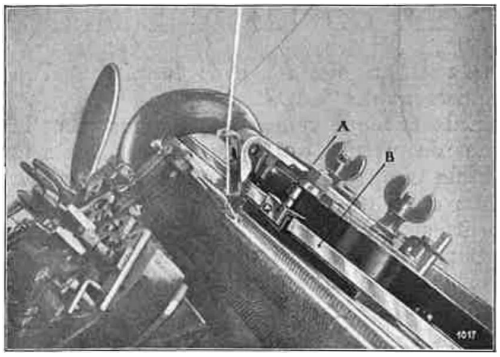


Fig. 62. Yarn carrier brake of the M machine.

Type MR machine.

The MR machine is similar in appearance to the MM machine, except that it has on the front, besides the usual stitch cams, adjustable auxiliary stitch cams, whose use is given on page 61, and that it is made in different widths and gauges.

The 60 cm MR machine is supplied with the MM table, the 80 cm one with the same table as the CS machine (see fig. 41), for the 80 cm MR machine is driven by a balance lever instead of a crank.

Unlike the CS machine, shown in fig. 41 with the cone holders at the sides, the 80 cm MR machine takes a cone holder like that of the MM machine (fig. 10).

The MR machine can take the plating tip and the push-button yarn changer. In gauges 25, 28, 32 and 36 it can also take the Super-8.

Type CS machine.

The type CS machine, shown in fig. 41, is a machine 80 cm wide or more, in different gauges. It is driven by a balance lever instead of a crank.

The carriage with its large bridge can pass over the yarn carriers, of which there are 2, one on the front and one on the rear. This makes it possible to change yarn carrier either on the right or on the left, or to knit 2 pieces at once, stopping the yarn carrier with a stop block at any point of the carriage's travel.

On a CS machine one can knit for example 2 legs of a pair of drawers at the same time, 2 sleeves, or the two sides of a neck opening. One is thus certain that the 2 pieces are exactly the same length.

The machine as normally supplied, with 2 yarn carriers and 4 stop blocks, makes it possible to knit 2 pieces at once if each of these pieces is in a single colour. To knit 2 pieces in 2 colours, on the other hand, one must of course obtain 2 yarn carriers and 4 stop blocks as well as 2 additional tension units.

The cones are placed at the 2 ends of the machine and not behind it as on the MM machine.

The CS machine is fitted with auxiliary stitch cams on the front (see their use on page 61).

The CS machine can take the plating tip and the push-button yarn changer; in the latter case, a special cone holder for 6 cones on the right of the machine is supplied with the changer (see fig. 41).

The CS machine in gauges 25, 28, 32 and 36 can also take the Super-8.

Auxiliary cams.

These cams, when not in use, are always fixed as high as possible.

Using the auxiliary stitch cams.

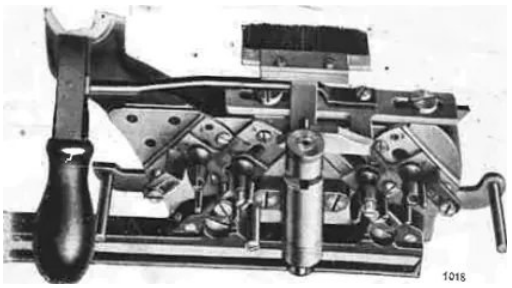


Fig. 63. Cam box with auxiliary stitch cams seen from above.

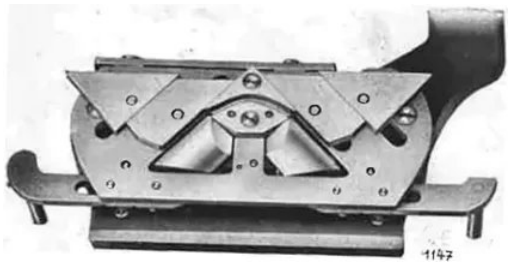


Fig. 64. Cam box with auxiliary stitch cams seen from below.

These two cams are triangular in shape and are in the front cam box, one on the right, the other on the left (see fig. 63).

The right-hand one bears No. 5, the left-hand one No. 6. They are used only for ordinary 1 × 1 rib and half cardigan; they allow these two kinds of knitting to be made with very fine, very even stitches. They are used preferably for high-quality articles or for certain 1 × 1 rib articles with a part tighter than the rest of the knitting (waistband, cuff, etc.).

Note that for knitting 1 × 1 rib with auxiliary stitch cams, a yarn a little finer than for ordinary 1 × 1 rib must be chosen, and of very good quality.

To knit with the auxiliary stitch cams, proceed as follows:

1 × 1 rib.

Cast on, knit the welt and stop the carriage on the left.

Pull out pins 1 and 3 and set the knobs as follows:

gauge 36 (No. 7):				gauge 32 (No. 8):			
Knobs	3	4	6	Knobs	3	4	6
at	16	3	15	at	15	3	14

Bring the carriage to the right and set the knobs as follows:

gauge 36 (No. 7):				gauge 32 (No. 8):			
Knobs	1	2	5	Knobs	1	2	5
at	16	3	15	at	15	3	14

Important notes:

For this kind of knitting it is essential to work with the rear stitch cams No. 2 and 3 free.

When another knob setting is used, stitch cams No. 1 and 4 must stay at position 3, and the rear stitch cams No. 2 and 3 *one* division lower than the auxiliary stitch cams No. 5 and 6.

Half cardigan.

Cast on, knit the welt and stop the carriage on the left. Pull out pins 1 and 3 and set the knobs as follows:

gauge 36 (No. 7):				gauge 32 (No. 8):			
Knobs	3	4	6	Knobs	3	4	6
at	16	3	15	at	15	3	14

Bring the carriage to the right and set the knobs as follows:

gauge 36 (No. 7):				gauge 32 (No. 8):			
Knobs	1	2	5	Knobs	1	2	5
at	16	3	10	at	15	3	10

Note:

For this kind of knitting it is essential to work with stitch cams No. 1 and 3 free. Stitch cam No. 3 must always be set *one* division lower than auxiliary stitch cam No. 6.

Racking attachment for the Super-8.

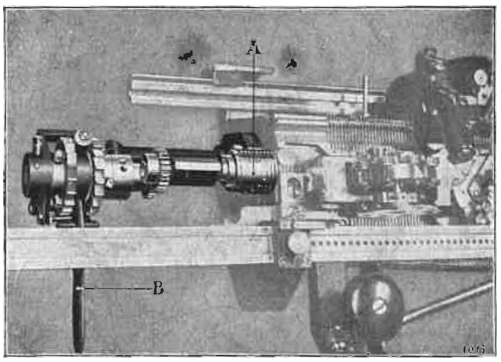


Fig. 65. Super-8 racking attachment.

This attachment, which can be fitted to all Dubied machines with the Super-8, is fixed to the left-hand wall of the frame. It shifts the starting position of the Super-8 selector cams, thus giving original effects such as zigzags, chevrons and staggered motifs.

The essential part of the attachment is a drum A, milled with stepped grooves. There are 9 of these grooves in gauges 32 and 36 and 12 in gauges 25 and 28. The spacing of 2 grooves for each gauge is the same as that of the needles on the bed.

With a lever B, the drum can be turned so as to bring a greater or smaller number of grooves into line with the bed, at the point where the toothed pinion (small toothed wheel) of the Super-8 selector cam in work engages.

The Super-8 racking attachment can be put in action only when the carriage is fully to the left.

Using the attachment.

Before starting to knit, check that the drum A, fig. 65, is in the out-of-work position. This position is recognised by the fact that no groove of the drum A lies in the path of the small toothed wheels of the Super-8. To obtain this out-of-work position, one need only pull the lever of the attachment towards oneself until the drum locks.

Begin knitting and put the Super-8 in action in the usual way.

The indicator strips for racked Super-8 patterns bear figures appearing in window 3, showing the position of the drum of the racking attachment. As the pattern was begun with the drum at position 0, the racking attachment is worked only when a figure appears in window 3. If this figure is for example 5, the lever of the racking attachment must be pressed fully 5 times. At each press the drum makes a fraction of a turn, which brings one more groove into the path of the toothed wheel of the selector cam.

Continue in the same way, taking care always to work the lever of the racking attachment fully home.

To go from positions 0 to 9, press the lever; to go from positions 9 to 0, pull the lever.